



2025 FRM Part II

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信用风险测量与管理

2025 年 3 月

2. Credit Risk Measurement and Management

2.1. Key Point: Credit Risk

2.1.1. 重要知识点

2.1.1.1. Define and Explain Credit Risk

- Credit risk refers to the possibility of loss of funds due to the inability, unwillingness or failure of a counterparty to meet its financial obligations in a timely manner.

2.1.1.2. Distinguish between Insolvency, Default and Bankruptcy

- Insolvency describes the financial state of a debtor whose liabilities exceed their assets.
- Default refers to the failure to meet an obligation under a contract, such as a failure to repay a debt on time.
- Bankruptcy occurs when a company files for Chapter 11 or Chapter 7 protection under the U.S. Bankruptcy Code and defaults on its obligations. During bankruptcy, the court reviews the financial position of the defaulting entity and negotiates with management, creditors and sometimes equity owners in an attempt to maintain the business by selling assets and/or renegotiating financing arrangements with lenders.

2.1.1.3. Identify and Describe Transactions that Create Credit Risk

- Understand the various types of transactions that can lead to credit risk, including but not limited to lending, leases, receivables and prepayment, asset claims, derivatives, etc., and point out that banks.

2.1.1.4. Entities Exposed to Credit Risk

- Describe entities (banks, asset managers, hedge funds, insurance companies, pension funds, corporate, and individuals) exposed to credit risk and the circumstances under which they are exposed.

2.1.1.5. Motivation to Manage and Take Credit Risk

- Emphasize the importance of managing credit risk for the survival, profitability and return on equity of all companies.

2.1.2. 基础题

Q-1. Which of the following statements accurately describes the distinction between insolvency, default, and bankruptcy?

- A. Insolvency occurs when a company fails to fulfill its contractual obligations, leading to a legal process aimed at restructuring its debts and assets.
- B. Default signifies a company's financial state where its assets exceed its liabilities, prompting a court intervention to dissolve the business.
- C. Bankruptcy refers to the breach of a contractual agreement due to temporary illiquidity, often leading to the liquidation of assets to meet payment obligations.
- D. Insolvency indicates a situation where a company's liabilities surpass its assets, while

default involves the failure to meet specific contractual obligations, and bankruptcy is a legal process addressing default situations.

Q-2. Which of the following statements accurately describes credit risk exposure in modern financial transactions?

- A. Sophisticated products, such as derivatives, may create significant credit risk exposure without involving immediate cash exchanges, necessitating thorough risk analysis and hedging strategies to minimize exposure.
- B. Credit risk primarily stems from traditional lending practices, such as corporate bonds and consumer loans, where the risk lies in nonrepayment, slow repayment, disputes, and other frictional costs.
- C. Leasing transactions, while involving potential nonpayment and costs associated with recovering leased assets, pose minimal credit risk due to the guaranteed insurance provided by insuring institutions.
- D. Deposits in banks are entirely risk-free, as they typically offer full insurance coverage, ensuring no exposure to credit risk for depositors.

Q-3. John is a risk manager for ABC, an insurance company. Below is a conversation between John and his colleague David about credit risk. John tells David, "ABC faces significant credit risk in our investment portfolio. We are taking on too much risk in emerging markets debt and need to find a way to hedge this risk." David responds, "I am more concerned about the credit risk we take on when originating new policies, in particular when we extend installment terms and grace periods to our corporate customers." John responds, "No, David, what we should worry about is our emerging market credit risk and the credit risk we are taking on from buying reinsurance." David responds, "Reinsurance? Reinsurance does not create credit risk. Reinsurance helps us mitigate our risk exposure." In the above conversation, each of the statements is likely true except which of the statements is most likely incorrect?

- A. John's statement about emerging market debt.
- B. David's statement about originating new policies.
- C. John's statement about reinsurance.
- D. David's statement about reinsurance.

2.2. Key Point: Credit VaR

2.2.1. 重要知识点

2.2.1.1. Credit VaR

Type of Credit Loss	Description	Formula
Expected Credit Loss	Represent the expected value of the credit loss	$EL = PD \times LGD \times EAD$
Unexpected Credit Loss	Quantile of the credit loss in excess of the expected loss	$UL = \text{Credit VaR} = WCL - EL$

2.2.1.2. Vasicek's model, Credit Risk Plus (Credit Risk+) model, and the Credit Metrics

- Vasicek's model employs a one-factor Gaussian copula to analyze default rate distribution and correlation for regulatory capital requirements.
- Credit Risk Plus focuses on individual firm default probabilities without explicitly modeling default correlation.
- Credit Metrics, on the other hand, incorporates both default and downgrade risks through a Monte Carlo simulation, providing a comprehensive approach to credit risk assessment.

2.1.1. 基础题

Q-4. A credit analyst at a bank has been asked to produce an exposure analysis for three of the loans in the bank's portfolio. Loan information assembled by the analyst as well as the bank's internal default.

Loan	Tenor (Years)	Notional (USD)	Loss Given Default	S&P Rating
1	2	30,000,000	0.75	BB-
2	3	100,000,000	0.90	A
3	1	100,000,000	0.70	B+

Probability of Default (PD)			
	Tenor (Years)		
Loan Quality	1	2	3
Investment Grade	0.01	0.02	0.03
Non-Investment Grade	0.05	0.10	0.20

There is no collateral provided by the borrower for these loans, so the analyst uses the notional amount provided above as the Exposure at Default. Which of the following correctly orders the expected loss for each loan from lowest to highest?

- A. Loan 1 < Loan 2 < Loan 3
- B. Loan 1 < Loan 3 < Loan 2
- C. Loan 2 < Loan 3 < Loan 1
- D. Loan 2 < Loan 1 < Loan 3

Q-5. A manager of a mutual fund has taken significant credit exposure to Europe and Asia. Concerned with uncertain market conditions, the manager wants to change the assumptions in the fund's risk models by increasing the default correlation between bonds issued in Europe and bonds issued in Asia. If the default correlation is increased and all the other parameters are kept the same, which of the following is true?

- A. The expected loss of the portfolio will increase.
- B. The unexpected loss of the portfolio will decrease.
- C. The expected loss of the portfolio will decrease.
- D. The unexpected loss of the portfolio will increase.

Q-6. The default correlation under a single-factor credit model is 4.9%. Both credits have the same individual default probabilities of 2%. The joint default probability is characterized by a bivariate standard normal distribution. Below listed the asset correlations implied by various joint default probabilities. What is the implied asset correlation?

Asset Correlation	Joint Default Probability
-	0.040%
0.05	0.053%
0.10	0.069%
0.15	0.040%
0.20	0.110%
0.25	0.136%

- A. 0.1
- B. 0.15
- C. 0.2
- D. 0.25

Q-7. Suppose there is a \$1,000,000 portfolio with $n = 50$ credits that each has a default probability of $\pi = 0.02$ and a zero recovery rate, the default correlation is 0. In addition, each credit is equally weighted and has a terminal value of \$20,000 if there is no default. The number of defaults is binomially distributed with parameters of $n = 50$ and $\pi = 0.02$, and the 95th percentile of the number of defaults based on this distribution is 3. What is the credit VaR at the 95% confidence level based on these parameters?

- A. \$30,000
- B. \$40,000
- C. \$50,000
- D. \$60,000

Q-8. Becky the Risk Analyst is trying to estimate the credit value at risk (CVaR) of a three-bond portfolio, where the CVaR is defined as the maximum unexpected loss at 99.0% confidence over a one-month horizon. The bonds are independent (i.e., no default correlation) and identical with a one-month forward value of \$1.0 million each, a one-year cumulative default probability of 4.0%, and an assumed zero recovery rate. Which is nearest to the one-month 99.0% CVaR?

- A. \$989,812
- B. \$1.0 million
- C. \$1.7 million
- D. \$2.3 million

Q-9. At the beginning of the year, a firm bought an AA-rated corporate bond at USD 110 per USD 100 face value. Using market data, the risk manager estimates the following year-end values for the bond based on interest rate simulations informed by the economics team:

Rating	Year-End Bonds Value
AAA	112
AA	109
A	105
BBB	101
BB	92
B	83
CCC	73
Default	50

In addition, the risk manager estimates the 1-year transition probabilities on the AA-rated corporate bond:

Rating	Probability of State
AAA	3.00%
AA	85.00%
A	7.00%
BBB	4.00%
BB	0.35%

B	0.25%
CCC	0.15%
Default	0.25%

What is the 1-year 95% credit VaR per USD 100 of face value closest to?

- A. USD 9
- B. USD 18
- C. USD 30
- D. USD 36

Q-10. Suppose there is a \$1,000,000 portfolio with n credits that each have a default probability, $\pi = 2\%$ and a zero recovery rate. The default correlation is 0 and $n = 1,000$. There is a probability of 28 defaults at the 95th percentile based on the binomial distribution with the parameters of $n = 1,000$ and $\pi = 0.02$. What is the credit VaR at the 95% confidence level based on these parameters?

- A. \$7,000
- B. \$8,000
- C. \$9,000
- D. \$10,000

Q-11. Consider a pair of two speculative credits, rated BB and BB-, with default probabilities respectively of 2% and 3%. If their joint default probability is 0.4%, which is nearest to the implied default correlation?

- A. Zero
- B. 0.083
- C. 0.1424
- D. 0.3750

Q-12. A credit analyst at a bank is asked to estimate the credit VaR (CVaR) for three loans in the bank's credit portfolio. The following tables are about loan information and the annual probability of default (PD).

Loan	Maturity (years)	Exposure (\$GD)	Loss given default	S&P rating
S	2	55,000,000	0.8	BBB
T	3	36,000,000	0.9	BB-
U	4	50,000,000	0.7	A

Loan maturity (years)	2	3	4
PD (investment grade)	1.5%	2.5%	3.5%
PD (non-investment grade)	5.0%	12.0%	18.0%

Assuming the 95th percentile of the WCL for the three loans are the same, which of the following is correct about the comparison of the 95% CVaR of the loans?

- A. CVaR of Loan S > CVaR of Loan U > CVaR of Loan T
- B. CVaR of Loan T > CVaR of Loan U > CVaR of Loan S
- C. CVaR of Loan T > CVaR of Loan S > CVaR of Loan U
- D. CVaR of Loan U > CVaR of Loan S > CVaR of Loan T

Q-13. Which method among the following is most concerned with estimating default losses by focusing solely on individual firm default probabilities without explicitly modeling default correlation?

- A. Vasicek's Model employs a one-factor Gaussian copula to approximate the expected loss distribution by considering default correlation within a portfolio over a specific time period.
- B. Credit Risk Plus (Credit Risk+) utilizes assumptions about the default probabilities of individual firms, similar to procedures found in the insurance industry, to calculate default losses.
- C. Credit Metrics integrates both default probabilities and credit migrations into its risk assessment, utilizing a Gaussian copula model combined with a rating transition matrix to project future credit ratings and assess losses from both downgrades and defaults.
- D. Gordy's extension of Vasicek's model considers a one-factor world assumption for larger portfolios to estimate regulatory capital requirements.

2.2. Key Point: Capital Structure

2.2.1. 重要知识点

2.2.1.1. EL and UL (in statistical terms)

$$EL = PD \times EA \times LR$$

$$UL = EA \times \sqrt{PD \times \sigma_{LR}^2 + LR^2 \times \sigma_{PD}^2}$$

- Where σ_{LR} = standard deviation of the loss rate LR
- σ_{PD} = standard deviation of the default probability PD
- $\sigma_{PD}^2 = PD(1 - PD)$

2.2.1.2. Unexpected Loss Contribution

$$ULMC_i = \frac{\partial UL_P}{\partial UL_i} = \frac{1}{2UL_P} \times \frac{\partial (UL_P^2)}{\partial UL_i}$$

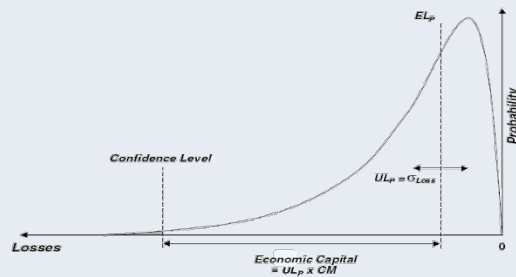
$$= \frac{1}{2UL_P} \times \frac{\partial \left(\sum_{i=1}^n \sum_{j=1}^n \rho_{ij} UL_i UL_j \right)}{\partial UL_i} = \frac{\sum_{j=1}^n UL_j \rho_{ij}}{UL_P}$$

2.2.1.3. Total Contribution to the Portfolio's UL

$$UL_P = \sum_{i=1}^n ULMC_i \times UL_i$$

$$ULC_i = ULMC_i \times UL_i = \frac{\sum_{j=1}^n UL_j \times \rho_{ij}}{UL_P} \times UL_i$$

2.2.1.4. Economic Capital



- Economic Capital_P = UL_P × CM
- Economic Capital_i = ULC_i × CM
- CM = capital multiplier

2.2.2. 基础题

Q-14. A bank's credit exposure to a customer consists of the following:

- Exposure amount is \$50 million
- Probability of default is 2%
- Loss rate is 50%
- Standard deviation of loss rate is 40%

Which is nearest to the exposure's unexpected loss?

- A. \$2.48 million
- B. \$3.29 million
- C. \$4.50 million
- D. \$7.75 million

Q-15. A bank has extended two loans to customers in the same industry. Both loans are have an exposure amount of \$50 million, default probability of 2%, loss rate of 50%, and standard deviation of loss rate of 60% such that each loan has an expected loss of

\$500,000 and an unexpected loss of \$5.5 million. In this way, the bank's credit portfolio consists of these two credit assets; and the default correlation between the two loans is 28%. Which is nearest to the risk contribution of each asset to the portfolio's unexpected loss?

- A. \$3.33 million
- B. \$4.40 million
- C. \$5.37 million
- D. \$5.50 million

Q-16. A risk analyst at a credit ratings agency is evaluating the economic capital for credit risk of two competing regional banks, Bank ABC and Bank XYZ. The two banks have the same credit asset exposure, duration of credit exposure, credit rating, and expected loss. Assuming the average pairwise default correlation between credit assets of Bank ABC is lower than that of Bank XYZ, and the two banks assess their risk appetite at the same predetermined confidence level, which of the following statements would be correct?

- A. If the confidence level (CI) for both banks is increased, the level of economic capital needed for Bank ABC and for Bank XYZ will both increase.
- B. If the CI for both banks is decreased, the level of economic capital needed will increase for Bank ABC but will decrease for Bank XYZ.
- C. If the CI for both banks is increased, the level of economic capital needed will decrease for Bank ABC but will increase for Bank XYZ.
- D. If the CI for both banks is kept unchanged, the level of economic capital needed for Bank ABC and for Bank XYZ will be equal.

2.3. Key Point: Measurement Methods of PD

2.3.1. 重要知识点

2.3.1.1. Real-world vs. Risk-neutral Default Probabilities

- Real-world default probabilities reflect actual borrower default likelihood based on historical data and market conditions, while risk-neutral default probabilities are derived from theoretical models assuming a risk-free environment. Real-world probabilities are grounded in empirical evidence, whereas risk-neutral probabilities facilitate consistent pricing of financial instruments. Risk-neutral probabilities are suitable for valuing derivatives, while real-world probabilities are better for scenario analysis and stress testing in credit risk management.

2.3.1.2. Judgmental Approaches, Data-Driven Empirical Models and Financial Models

- Judgmental Approaches: Relies on expert judgment to assess credit risk based on qualitative and quantitative borrower attributes without direct mathematical formulae.

- Data-Driven Empirical Models: Utilize historical data to statistically estimate the probability of default and build relationships between risk factors and lending outcomes.
- Financial Models: Based on economic and financial principles. Employ theoretical frameworks and financial instruments' market data to quantify credit risk dynamically, such as the Merton model estimating default probability and distance to default.

2.3.2. 基础题

Q-17. What is the primary distinction between real-world default probabilities and risk-neutral default probabilities in credit risk analysis?

- A. Real-world probabilities are derived from theoretical models, while risk-neutral probabilities are grounded in empirical evidence and market observations.
- B. Real-world probabilities consider the inherent uncertainties of the financial market, whereas risk-neutral probabilities assume a risk-free environment without any additional return demands.
- C. Risk-neutral probabilities are more suitable for scenario analysis and stress testing, while real-world probabilities are appropriate for valuing financial instruments consistently with other market instruments.
- D. Risk-neutral probabilities are lower than real-world probabilities due to the expected risk premium, while real-world probabilities are higher as they align with arbitrage-free pricing principles.

Q-18. Which of the following statements best describes financial models used in predicting default?

- A. Financial models rely on expert judgment to evaluate qualitative and quantitative characteristics of borrowers.
- B. Financial models are constructed based on historical data related to loan outcomes and employ statistical techniques to establish relationships between default probability and input variables.
- C. Financial models adopt a normative approach rooted in fundamental economic principles and aim to describe the mechanics of the default process.
- D. Financial models utilize market data on bonds and credit derivatives to infer a firm's credit risk structure and tend to be more sophisticated, allowing for dynamic analysis and scenario testing.

2.4. Key Point: Probability of Default in Rating System

2.4.1. 重要知识点

2.4.1.1. External and Internal Credit Ratings

- External credit rating agencies, such as Moody's, S&P, and Fitch, provide standardized

assessments of creditworthiness for publicly traded corporate bonds. These ratings aim for stability and consistency across multiple issuers. In contrast, internal credit rating systems used by banks focus on customized methodologies tailored to evaluate specific clients, including smaller enterprises. These internal ratings react more promptly to market changes and inform lending decisions.

2.4.1.2. Through-the-Cycle and Point-in-Time Ratings

- Through-the-cycle (TTC) credit ratings provide a stable, long-term assessment of credit risk across economic cycles. In contrast, point-in-time (PIT) ratings focus on short-term outlooks and respond promptly to recent information, making them suitable for immediate credit risk evaluation.

2.4.1.3. Process of Developing Credit Risk Scoring and Rating Models

- The development of credit risk scoring and rating models involves gathering and preprocessing extensive borrower data, followed by model fitting and rigorous validation to ensure accurate prediction of creditworthiness.

2.4.1.4. Cumulative Default Probability

$$PD_k^{\text{cumulated}} = \frac{Def_i}{Names_t}$$

2.4.1.5. Marginal Default Probability

$$PD_k^{\text{marg}} = PD_{t+k}^{\text{cumulated}} - PD_t^{\text{cumulated}}$$

2.4.1.6. Forward Probability (Contingent to the Survival Rate)

$$PD_{t,t+k}^{\text{Forw}} = \frac{Def_{t+k} - Def_t}{Names \text{ survived}_t}$$

2.4.1.7. Survival Rate

$$SR_{t,t+k}^{\text{Forw}} = (1 - PD_{t,t+k}^{\text{Forw}}) (1 - PD_t^{\text{cumulated}}) = \prod_{i=1}^t SR_i^{\text{Forw}}$$

2.4.2. 基础题

Q-19. Which of the following statements about agencies' ratings and internal credit rating systems is true?

- Internal credit rating systems used by banks primarily rely on market information to estimate probabilities of default and aim for quick responsiveness to changes.
- Rating agencies like Moody's and S&P frequently change credit ratings to ensure stability and avoid frequent fluctuations in the market.
- Internal credit rating systems often focus on profitability ratios and balance sheet indicators to assess creditworthiness, recognizing the importance of cash flow in loan

12-90

repayment.

- D. Moody's KMV and Kamakura provide credit ratings for small and medium-sized companies that do not have publicly traded debt, ensuring comprehensive coverage in the market.

Q-20. Which of the following statements best distinguishes between through-the-cycle (TTC) and point-in-time (PIT) credit rating systems?

- A. Through-the-cycle ratings aim to provide a stable measure of credit risk over an extended period, while point-in-time ratings focus on short-term fluctuations and immediate credit risk.
- B. Through-the-cycle ratings are regularly updated to reflect the latest borrower status, while point-in-time ratings capture credit risk throughout an entire economic cycle.
- C. Through-the-cycle ratings emphasize the immediate credit risk faced by the borrower, while point-in-time ratings filter out short-term noise to offer a steady picture of creditworthiness.
- D. Through-the-cycle ratings are suitable for short-term transactions and loans, while point-in-time ratings provide a conservative and forward-looking assessment of credit risk.

Q-21. What is an essential aspect of the development process for credit risk scoring and rating models?

- A. The implementation phase focuses on mapping credit scores to risk rating classes based on empirical PD estimates and ensuring a well-diversified distribution of borrowers across rating categories.
- B. Model validation primarily involves comparing the credit model's results to those of external assessments, such as ratings issued by credit rating agencies, to identify significant discrepancies.
- C. During data collection and pre-processing, expert judgment is considered unnecessary, and all attribute selection is based solely on algorithmic procedures and statistical tests.
- D. The development process does not involve continuous evaluation of borrowers; instead, borrowers are rated once, and their ratings remain static over time.

Q-22. An analyst has noted that the default frequency in the pharmaceutical industry has been constant at 8% for an extended period of time. Based on this information, which of the following statements is most likely correct for a randomly selected firm following a Bernoulli distribution?

- I. The cumulative probability that a randomly selected firm in the pharmaceutical industry will default is constant.

II. The probability that the firm survives for the next 6 years without default is approximately 60%.

- A. I only
- B. II only
- C. Both I and II
- D. Neither I nor II

Q-23. A portfolio consists of 10 independent bonds. There will be one default on average in 5 years. What is the probability that only exactly one default in a year.

- A. 16.37%
- B. 26.84%
- C. 36.96%
- D. 45.28%

Q-24. As a result of the credit crunch, a small retail bank wants to better predict and model the likelihood that its larger commercial loans might default. It is developing an internal ratings-based approach to assess its commercial customers. Given this one-year transition matrix, what is the probability that a loan currently rated at B will default over a two-year period?

Rating at Beginning of Period	Rating at End of Period			
	A	B	C	D
A	0.90	0.10	0.00	0.00
B	0.00	0.75	0.15	0.10
C	0.00	0.05	0.55	0.40

- A. 17.50%
- B. 20.0%
- C. 21.1%
- D. 23.5%

Q-25. A risk analyst at a mid-size hedge fund is evaluating the credit risk of several trade positions. The hedge fund specializes in corporate debt and runs a strategy that utilizes both relative value and long-only trades using CDS and bonds. One of the new trades at the hedge fund is a B-rated long bond valued at JPY 10 billion. Some of the hedge fund's newest clients, including the B-rated bond holders, are restricted from withdrawing their funds for four years. The analyst is currently evaluating the impact of various default scenarios to estimate future asset liquidity. The analyst has estimated that the forward probability of default of the B-rated bond is 7.7% in Year 1; 7.1% in Year 2; 6.6% in Year

3; and 6.1% in Year 4. What is the probability that the bond survives for 3 years and then defaults during Year 4?

- A. 4.9%
- B. 5.7%
- C. 6.1%
- D. 6.9%

2.5. Key Point: Merton Model

2.5.1. 重要知识点

2.5.1.1. Merton Model

- Equity
 - Equity is a call option on the firm value with strike price equal to the face value of debt.
- Risky Bond
 - Value of Risky Debt = Risk-Free Bond - put on Firm
- Merton Model
 - $\text{Equity} = VN(d_1) - Ke^{-rT}N(d_2)$
 - $\text{Debt} = Ke^{-rT} - \text{put} = VN(-d_1) + Ke^{-rT}N(d_2)$
 - $d_{1,2} = \frac{\ln(V/Ke^{-rT})}{\sigma\sqrt{T}} \pm \frac{\sigma\sqrt{T}}{2}$
 - V: firm value
 - K: face value of debt
- Risk-Neutral vs. Real World

2.5.2. 基础题

Q-26. A firm has an asset value of \$110 million with asset volatility of 30% per annum. Its only debt is a zero-coupon bond with face value of \$80 million that matures in five years. The risk-free rate is 4%. The Black-Scholes Merton price of a put option on the firm's assets with strike price equal to the face value of the bond is \$6.95 million. Which is near to the current value of the firm's debt.

- A. \$6.95 million
- B. \$41.30 million
- C. \$58.55 million
- D. \$65.50 million

Q-27. Suppose a firm has two debt issues outstanding. One is a senior debt issue that matures in three years with a principal amount of \$100 million. The other is a subordinate debt

issue that also matures in three years with a principal amount of \$50 million. The annual interest rate is 5%, and the volatility of the firm value is estimated to be 15%. If interest rates decline in the Merton model, then which of the following is true?

- A. If the firm is experiencing financial distress (low firm value), then the value of senior debt will increase while the values of subordinate debt and equity will both decline.
- B. If the firm is not experiencing financial distress (high firm value), then the value of senior debt and subordinate debt and equity will increase.
- C. If the firm is experiencing financial distress (low firm value), then the value of senior debt and subordinate debt will increase while equity values will decline.
- D. If the firm is not experiencing financial distress (high firm value), then the value of senior debt will increase while the values of subordinate debt and equity will both decline.

Q-28. In the following things about Merton model, which of the statement is true?

- A. In Merton model the payment to debt holder can be seen as the payoff of a riskless bond plus a put on the value of the firm.
- B. The sudden surprise (a jump), leading to an unexpected default can be captured by the by this model.
- C. The model can take into account the default prior to the maturity of debt, when a borrower claims so.
- D. The value of the firm is difficult to pin down cause the market-to-market value of debt is often unknown.

Q-29. Consider a firm with current asset value of \$20 billion, asset volatility of 35% per annum, short-term liabilities of \$12 billion and long-term liabilities of \$6 billion. The expected return on the firm's assets is 12% and the risk-free rate is 1%. Finally, the firm does not pay dividends and the credit horizon is 1 year. If the strike price default point is the sum of short-term debt plus one-half of long-term debt, what is the Merton physical probability of default in one year?

- A. 10.11%
- B. 12.11%
- C. 14.11%
- D. 16.11%

z	P(Z < z)	z	P(Z < z)	z	P(Z < z)
-1.50	0.0668	-1.00	0.1587	-0.50	0.3085
-1.49	0.0681	-0.99	0.1611	-0.49	0.3121
-1.48	0.0694	-0.98	0.1635	-0.48	0.3156
-1.47	0.0708	-0.97	0.1660	-0.47	0.3192

-1.46	0.0721	-0.96	0.1685	-0.46	0.3228
-1.45	0.0735	-0.95	0.1711	-0.45	0.3264
-1.44	0.0749	-0.94	0.1736	-0.44	0.3300
-1.43	0.0764	-0.93	0.1762	-0.43	0.3336
-1.42	0.0778	-0.92	0.1788	-0.42	0.3372

Q-30. The capital structure of HighGear Corporation consists of two parts: one 5-year zero-coupon bond with a face value of USD 100 million and the rest is equity. The current market value of the firm's asset (MVA) is USD 130 million and the risk-free rate is 25%. The firm's assets have an annual volatility of 30%. Assume that firm value is log-normally distributed with constant volatility. The firm's risk management division estimates the distance to default (in terms of number of standard deviations) using the Merton Model, Given the distance to default, the estimated risk-neutral default probability is $(N(1.9191) = 0.9724)$:

- A. 2.74%
- B. 12.78%
- C. 12.79%
- D. 30.56%

Q-31. A firm has a value of \$400 million with expected return of 14% per annum and volatility of 36% per annum. The firm's only debt is a short-term zero-coupon bond with face value of \$300 million due in one year. The riskless rate is 4%. Which is nearest to the firm's (normal returns-based) distance to default when deriving the physical PD?

- A. 1.0
- B. 2.7
- C. 3.3
- D. 8.5

Q-32. A credit risk analyst has estimated the probability of a particular firm defaulting in the next year to be 1.25% using the Merton model. The risk analyst used his bank's definition of the default threshold, namely that default occurs when the firm's value falls below the value of its short term debt plus half the value of its long term debt. Suppose the bank switched from using the Merton model to using the KMV approach to estimate default risk with the following historical expected default frequency buckets ($N(-2.24) = 1.25\%$):

Distance-to-Default	Expected Default Frequency
< -4	0.3%
-4 to -3	0.3%

-3 to -2.5	0.6%
-2.5 to -2.0	1.6%
-2.0 to -1.6	3.8%
-1.6 to -1.2	8.3%
-1.2 to -0.9	14.9%
-0.9 to -0.6	22.7%

What would the new default probability be?

- A. 0.3%
- B. 1.6%
- C. 2.8%
- D. 3.8%

Q-33. Analyst Greg is employing the Merton model to both value a firm's equity and estimate a physical default probability. He has collected the following information:

- The firm's default threshold one year forward is \$10 million; e.g., face value of short-term debt is \$10 million.
- The firm current asset value is \$12.75 million with an expected return of 8% per annum with continuous compounding
- The volatility of the firm's assets is 9.6%
- The risk-free rate is 2%

His exercise includes two components: one, valuation of the firm's equity market value by treating equity as a call option on the firm's assets; two, estimate of default probability by calculation of a forward distance to default. Greg makes two assumptions:

- I. An increase in the risk-free rate will increase an estimate of the firm's current equity market value, and
- II. An increase in the risk-free rate will decrease the estimated default probability.

Which of Greg's two assumptions is correct?

- A. Neither
- B. I only
- C. II only
- D. Both

Q-34. A credit manager in the counterparty risk division of a large bank uses a simplified version of the Merton model to monitor the relative vulnerability of its largest counterparties to changes in their valuation and financial conditions. In order to assess the risk of default of three particular counterparties, the manager calculates the distance to default

assuming a 1-year horizon ($t = 1$). The counterparties: Company P, Company Q, and Company R, belong to the same industry. Selected information on the companies is provided in the table:

Company	P	Q	R
Market value of asset (EUR million)	100	150	250
Face value of debt (EUR million)	60	100	160
Annual volatility of asset values	10.0%	7.0%	8.0%

Using the information above with the assumption that short-term debt is the only liability for each company, and the approximation formula of the distance to default, what is the correct ranking of the counterparties, from most likely to least likely to default?

- A. P; R; Q
- B. Q; P; R
- C. Q; R; P
- D. R; Q; P

Q-35. In the Merton model for credit risk, a firm's equity is treated as a call option on its assets.

Assume the following parameters are given:

- Equity value (E) = \$50 million
- Equity volatility (σ_E) = 40%
- Debt face value (D) = \$80 million
- Risk-free rate (r) = 5%
- Time to maturity (T) = 2 years
- $N(d_1) = 0.7$ and $N(d_2) = 0.4$, where d_1 and d_2 are defined in the Merton framework.

What are the firm's asset value (V) and asset volatility (σ_V)?

- A. $V = \$100$ million, $\sigma_V = 40\%$
- B. $V = \$112.8$ million, $\sigma_V = 25.3\%$
- C. $V = \$122.4$ million, $\sigma_V = 21.4\%$
- D. $V = \$130$ million, $\sigma_V = 32\%$

2.6. Key Point: Moody – KMV Model

2.6.1. 重要知识点

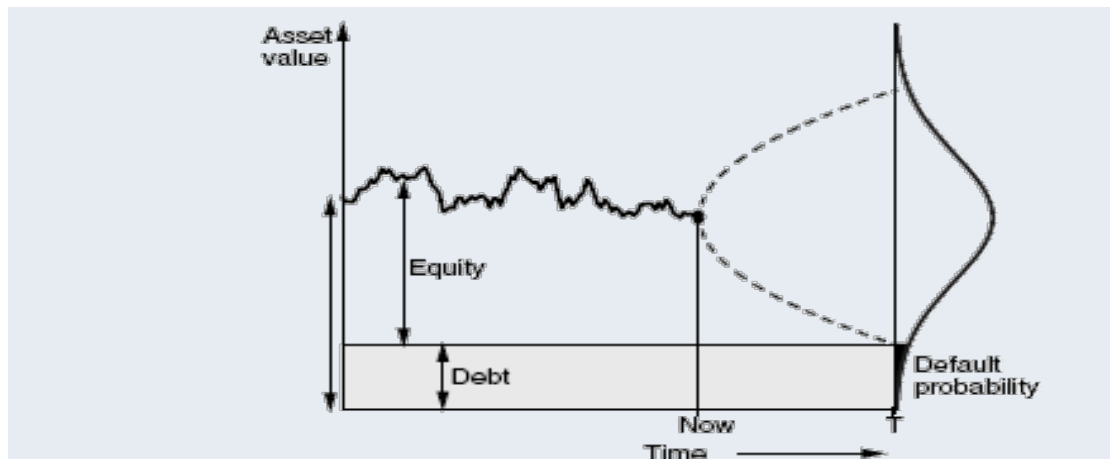
2.6.1.1. Moody – KMV Model

➤ The normalized distance to default (DD)

$$DD = \frac{A - K}{\sigma_A}$$

Where: K is the value of liabilities:

$K = \text{short-term liabilities} + 0.5 \times \text{long-term liabilities}.$



2.6.2. 基础题

Q-36. An analyst is using Moody's KMV model to estimate the distance to default of a large public firm, Shoos Inc., a firm that designs, manufactures and sells athletic shoes. The firm's capital structure consists of USD 40 million in short-term debt, USD 20 million in long-term debt, and there are one million shares of stock currently trading at USD 10 per share. The asset volatility is 20% per year. What is the normalized distance to default for Shoos Inc.?

- A. 0.714
- B. 1.430
- C. 2.240
- D. 5.000

Q-37. You are given the following information about firm A:

- Market value of asset at time 0 = 1000
- Market value of asset at time 1 = 1200
- Short-term debt = 500
- Long-term debt = 300
- Annualized asset volatility = 10%

According to KMV model, what are the default point and the distance to default at time 1?

	Default Point	Distance to Default
A.	800	3.33
B.	650	7.50
C.	650	4.58
D.	500	5.83

Q-38. Each of the following is true about the KMV model except which is false?

20-90

- A. Unlike Merton, which assume the default threshold is total debt, KMV's default threshold falls between short-term and total (short-term + long-term) debt.
- B. Similar to the Merton model, the KMV approach requires an estimate of asset volatility and future asset value in order to calculate distance to default as a number of standard deviations.
- C. Similar to the Merton model, the KMV approach models distance to default (DD) = (asset market value – default threshold)/(asset market value × asset volatility)
- D. Similar to the Merton model, the KMV approach assumes the future asset value is lognormal such that asset (log) returns are normal with EDF = N(DD)

2.7. Key Point: Hazard Rate

2.7.1. 重要知识点

2.7.1.1. Hazard Rates

- Also known as default intensity
- denoted as $\lambda(t)$
- Measure of conditional default probability in a short time period Δt

2.7.1.2. Cumulative PD

$$Q(t) = 1 - e^{-\bar{\lambda}(t)t}$$

2.7.1.3. CDS Spread and Hazard Rates

$$\bar{\lambda} = \frac{s(T)}{1 - R}$$

2.7.2. 基础题

Q-39. Given a hazard rate of 0.15, find the probability when a company defaults in year two given surviving the first year.

- A. 0.1393
- B. 0.2592
- C. 0.7408
- D. 0.8607

Q-40. Peter the municipal bond analyst observes that in recent years there have occurred only about 6 U.S. municipal defaults per year. If he makes the highly simplifying assumption that 6 defaults per year is the average in a Poisson process (distribution), what is the probability that the next municipal default will occur within one month?

- A. 8.42%
- B. 17%
- C. 39.35%

D. 60.65%

- Q-41.** Suppose the hazard rate is constant and equal to 0.090. In this case, each of the following is true except which is false?
- A. The unconditional one-year default probability is 8.6%
 - B. The unconditional two-year default probability is 16.5%
 - C. The probability of joint event of survival through the first year and default in the second year is 7.9%
 - D. The conditional one-year default probability, given survival through the first year, is 17.3%
- Q-42.** An analyst estimates that the hazard rate for a company is 0.1 per year. The probability of survival in the first year followed by a default in the second year is closest to:
- A. 8.61%
 - B. 9.00%
 - C. 9.52%
 - D. 19.03%

2.8. Key Point: Single – Factor Model

2.8.1. 重要知识点

2.8.1.1. Single – Factor Model

$$\alpha = \beta m + \sqrt{1 - \beta^2} \varepsilon$$

$$E(\alpha) = 0; \text{Var}(\alpha) = \beta^2 + 1 - \beta^2 = 1$$

➤ Conditional cumulative default probability function:

$$p = \Phi \left(\frac{K_i - \beta_i \bar{m}}{\sqrt{1 - \beta_i^2}} \right) \quad i = 1, 2, \dots$$

2.8.2. 基础题

- Q-43.** Under single-factor model, a firm has a beta of 0.40 and an unconditional default probability of 1%. If we enter a modest economic downturn, such that the value of $m = -1.0$, what is the conditional default probability?
- A. 1.0%
 - B. 1.8%
 - C. 2.5%
 - D. 2.8%
- Q-44.** Suppose an analyst use the single-factor model to measure portfolio credit risk, he starts

by imagining a number of firms $i = 1, 2, \dots$, each with its own correlation β_i to the market factor. Assume firm 1 is "cyclical" and has $\beta_1 = 0.5$, while firm 2 is "defensive" and has $\beta_2 = 0.1$. What is the asset return correlation of the two firms?

- A. 0.5
- B. 0.1
- C. 0.05
- D. 0.22

2.9. Key Point: Z-Score Model

2.9.1. 重要知识点

2.9.1.1. Altman's Z-Score Model

$$Z = 1.2X_1 + 1.40X_2 + 3.30X_3 + 0.6X_4 + 0.999X_5$$

x_1 is working capital/total assets

x_2 is retained earnings/total assets

x_3 is EBIT/total assets

x_4 is equity market value/book value of total liabilities,

x_5 is sales/total assets, and z is a number.

Z-Score	Interpretation
Greater than 3.0	Unlikely to default
Between 2.7 and 3.0	On alert
Between 1.8 and 2.7	Good chance of default
Less than 1.8	Very high probability of financial embarrassment.

2.9.2. 基础题

Q-45. The following financial data pertains to ABC Corp:

Working capital	\$170,000
Total assets	\$670,000
Book value of liabilities	\$240,000
Market value of liabilities	\$205,000
Retained earnings	\$300,000
Market value of equity	\$380,000
Earnings before interest and taxes	\$60,000
Sales	\$2,200,000

The Altman's Z-score equation is as follows:

$$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 0.999X_5$$

When using the computed Z-score to classify ABC based on credit quality, what is the

most appropriate classification for the company?

- A. Unlikely to default.
- B. On alert.
- C. Good chance of default.
- D. High likelihood of default.

2.10. Key Point: Credit Exposure

2.10.1. 重要知识点

2.10.1.1. Expected Positive Exposure (EPE)/Expected Negative Exposure (ENE)

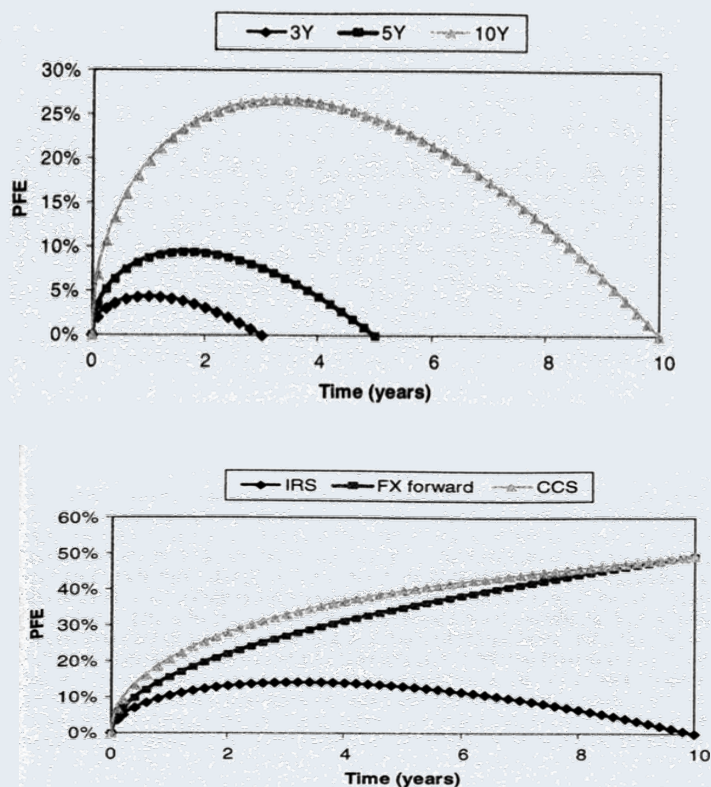
- Averages of the positive and negative exposures.
- Average across all values where negative or positive ones are set to zero.
- EPE is sometimes known as Expected Exposure (EE)
- ENE is sometimes known as Negative Expected Exposure (NEE) This represents the positive exposure from the counterparty's point of view.

2.10.1.2. Peak Exposure/Potential Future Exposure

- A high-percentile (typically 95% or 99%) of the distribution of exposures at any particular future date before the maturity date of the longest transaction in the netting set.

2.10.1.3. Maximum PFE

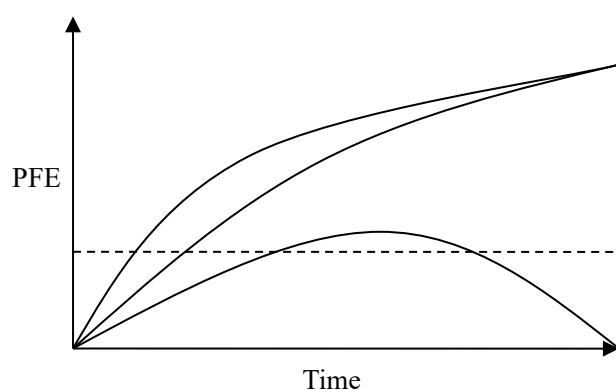
- Represents the highest PFE value over a given time interval, thus representing the worst-case exposure over the entire interval.



2.10.2. 基础题

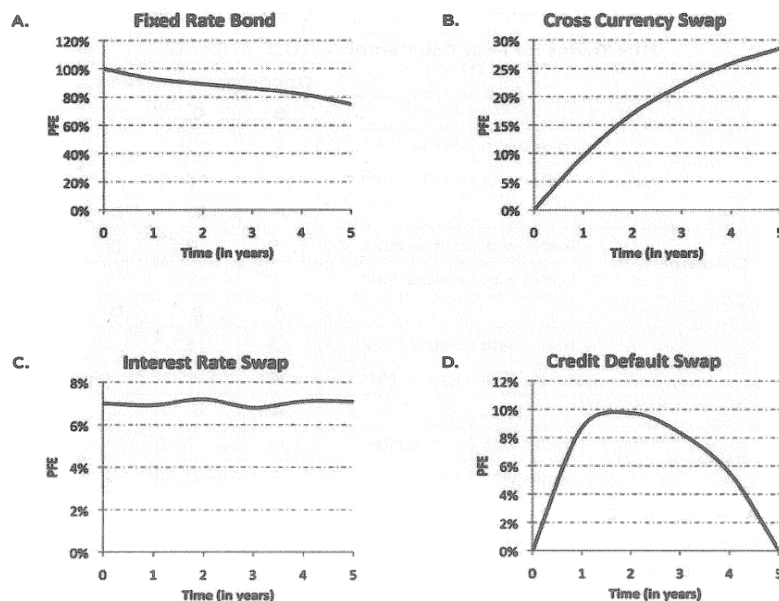
- Q-46.** Which of the following statements regarding exposure measures is not accurate?
- A. Assuming no recovery in bankruptcy, current exposure is the greater of zero and the market value of a transaction or portfolio of transactions that the investor would be lost when a counterparty defaulted.
 - B. Potential Future Exposure measures the distribution of exposures of low percentile.
 - C. Expected positive exposure measures averages of the positive exposures where negative ones are set to zero.
 - D. The weights of average expected positive exposure could represent the proportion that an individual expected positive exposures of the entire time interval.
- Q-47.** If a counterparty defaults before maturity, which of the following situations will cause a credit loss?
- A. You are short Euros in a one-year euro/USD forward FX contract, and the euro has appreciated.
 - B. You are short Euros in a one-year euro/USD forward FX contract, and the euro has depreciated.
 - C. You sold a one-year OTC euro call option, and the euro has appreciated.
 - D. You sold a one-year OTC euro call option, and the euro has depreciated.
- Q-48.** Paul sells a put option on HRTB stock with a time to expiration of six months, a strike price of USD 125, and underlying asset price of USD 98, implied volatility of 20% and a risk-free rate of 4%. What is Paul's counterparty credit exposure from this transaction?
- A. USD 0.00
 - B. USD 0.38
 - C. USD 1.75
 - D. USD 24.90
- Q-49.** Assume that swap rates are identical for all swap tenors. A swap dealer entered into a plain-vanilla swap one year ago as the receive-fixed party, when the price of the swap was 7%. Today, this swap dealer will face credit risk exposure from this swap only if the value of the swap for the dealer is
- A. Negative, which will occur if new swaps are being priced at 6%
 - B. Negative, which will occur if new swaps are being priced at 8%
 - C. Positive, which will occur if new swaps are being priced at 6%
 - D. Positive, which will occur if new swaps are being priced at 8%

- Q-50.** Which one of the following deals would have the greatest credit exposure for a \$1,000,000 deal size (assume the counterparty in each deal is an AAA-rated bank and has no settlement risk)?
- A. Pay fixed in an Australian dollar (AUD) interest rate swap for one year.
 - B. Sell USD against AUD in a one-year forward foreign exchange contract.
 - C. Sell a one-year AUD cap.
 - D. Purchase a one-year certificate of deposit.
- Q-51.** BNP Paribas has just entered into a plain-vanilla interest-rate swap as a pay-fixed counterparty. Credit Agricole is the receive-fixed counterparty in the same swap. The forward spot curve is upward-sloping. If LIBOR starts trending down and the forward spot curve flattens, the credit risk from the swap will:
- A. Increase only for BNP Paribas
 - B. Increase only for Credit Agricole
 - C. Decrease for both BNP Paribas and Credit Agricole
 - D. Increase for both BNP Paribas and Credit Agricole
- Q-52.** The chart below shows three exposure profiles, where the exposure metric is the potential future exposure (PFE): PFE of an interest rate swap (IRS), PFE of a foreign exchange (FX) forward contract, and PFE of a cross-currency swap. Also plotted is the average PFE of the interest rate swap, where "average PFE" is what Jorion calls the average worst credit exposure (AWCE). Which position's (instrument's) exposure profile is most likely the uppermost, concave plot line?



- A. PFE of interest rate swap
- B. PFE of foreign exchange (FX) contract
- C. PFE of cross-currency swap
- D. Average PFE of interest rate swap

Q-53. Which of the following graphs is an accurate representation of a typical potential future exposure (PFE) profile for the corresponding instrument?



Q-54. Six months ago, Textile Manufacturing Inc. (TMI) entered into a 9-month forward contract with Spin Mills Company (SMC) to purchase 36,000 tons of yarn. At the time the forward was entered into, 36,000 tons of yarn was priced at EUR 92.0 million but is currently priced at EUR 94.0 million. The continuously compounded risk-free rate has remained stable at 3.0% per year and is not expected to change during the entire contract period. Assuming the forward is fairly priced, what is the current potential credit risk exposure on the forward contract and who bears the risk?

- A. EUR 0.610 million; TMI bears the potential credit risk
- B. EUR 0.610 million; SMC bears the potential credit risk
- C. EUR 1.308 million; TMI bears the potential credit risk
- D. EUR 1.308 million; SMC bears the potential credit risk

2.11. Key Point: Credit Value Adjustment (CVA)

2.11.1. 重要知识点

2.11.1.1. Credit Value Adjustment (CVA)

- CVA is the expected value or price of counterparty credit risk. It represents a cost to the counterparty that bears a greater propensity to default.

$$CVA = -LGD \sum_{i=1}^m EPE(t_i) \times PD(t_{i-1}, t_i)$$

$$CVA = -\text{average EPE} \times \text{Spread}$$

- Incremental and Marginal CVA

- Incremental CVA calculates the cost of a new trade versus an existing one to

determine the effect that the new trade has on CVA. The formula is identical to stand-alone CVA, except for the use incremental expected positive exposure.

- Marginal CVA is used for trade level attribution. The formula is identical to stand-alone CVA, except for the use of marginal expected positive exposure.

➤ Stress Test on a Loan Portfolio

$$EL_s = \sum_{i=1}^N PD_i^s \cdot EAD_i \cdot LGD_i$$

$$EL_s - EL$$

➤ Stress Test on Derivative Portfolio

$$EL = \sum_{i=1}^N PD_i \cdot \alpha \cdot \text{average EPE}_i \cdot LGD_i$$

$$EL_s = \sum_{i=1}^N PD_i^s \cdot \alpha \cdot \text{average EPE}_i^s \cdot LGD_i$$

➤ Stressed CVA and Stress Loss on CVA

$$CVA^s = - \sum_{n=1}^N LGD_n \cdot \sum_{i=1}^m EPE_n^s(t_i) \cdot PD_n^s(t_{i-1}, t_i)$$

2.11.2. 基础题

Q-55. With respect to the CVA calculation, which of the following statement is correct when a risk manager wishes to understand which trades have the greatest impact on a counterparty's CVA? The manager would use:

- Incremental CVA because it accounts for the change in CVA once the new trade is priced, accounting for netting.
- Marginal CVA because he could break down netted trades into trade level contributions.
- Incremental CVA because he could break down netted trades into trade level contributions.
- Marginal CVA because it accounts for the change in CVA once the new trade is priced, accounting for netting.

Q-56. A risk manager needs a quick calculation of the BCVA on a swap. Assume inputs are as follows: average EPE = 5%, average ENE = -3%, counterparty credit spread = 300bps, financial institution credit spread = 200 bps. Compute BCVA from the perspective of the

financial institution.

- A. -1
- B. 1
- C. 9
- D. -9

Q-57. A bank enters into a swap agreement with a counterparty. The swap has no collateral requirements, and no netting agreements are present between the bank and the counterparty. The following data is available for the swap position:

- The counterparty expected positive exposure is 0.40% and approximately constant from month to month.
- The credit spread for a five-year credit default swap on the counterparty is 500 bps.
- The counterparty's probability of default within five years is 10%.
- The 5-year effective duration of the swap is 4.0.

Assuming no wrong-way risk on the position, which value is the closest approximation of the credit value adjustment expressed as a running spread?

- A. -2 bps
- B. -4 bps
- C. -5 bps
- D. -8 bps

Q-58. Mary assigns to John a long position in an at-the-money (ATM) call option with a one-year term and a strike price of \$100.00. The current stock price is \$100.00 with volatility of 60.0%. The risk-free rate is 3.0% with continuous compounding. $N(d_1) = 0.64$ and $N(d_2) = 0.40$. The present-valued expected positive exposure (EPE) to the counterparty, who holds the short option position, is \$23.00 with a probability of counterparty default of 5.0% and loss given default (LGD) of 75.0%.

Which is nearest to John's payment for the long option position, if his cost includes a credit valuation adjustment (CVA)?

- A. \$6.15
- B. \$19.37
- C. \$24.32
- D. \$26.04

Q-59. Sam prices a put option on an asset with the Black-Scholes-Merton option pricing model and calculates a model premium of \$25. This \$25 also coincidentally equals the present-valued expected positive exposure faced by Sam with respect to the short option position.

29-90

Sam estimates the probability of counterparty default by the option writer to be 10% with loss given default of 40%, such that the expected loss = \$25 EPE × 10% PD × 40% LGD = \$1.

He concludes that the CVA-adjusted (net of counterparty risk) option price is \$24. His colleague Jane observes that this calculation assumes no wrong-way risk. But there is a high, positive correlation between underlying asset price and the credit quality of the option writer counterparty: both the counterparty and underlying share a sector that reacts to the same common factors such that adverse economic regimes depress sector asset prices while lowering sector credit quality (and increasing credit spreads).

Is Jane correct that the CVA-adjusted option value deserves further adjustment?

- A. As the correlation is positive, this is instead right-way risk; but the true CVA-adjusted value remains \$24 as there is no adjustment for right-way risk.
- B. As the correlation is positive, this is instead right-way risk; therefore, the true CVA-adjusted value will be higher than \$24.
- C. Jane is correct that this is wrong-way risk; therefore, true CVA-adjusted value will be lower than \$24.
- D. Jane is correct that this is wrong-way risk but expected loss is not impacted by correlation, so Sam correctly has the CVA-adjusted value at \$24.

Q-60. Local Company, a frequent user of swaps, often enters into transactions with Global Bank, a major provider of swaps. Recently, Global Bank was downgraded from a rating of AA+ to a rating of A, while Local Company was downgraded from a rating of A to a rating of A-. During this time, the credit spread for Global Bank increased from 20 bps to 150 bps. While the credit spread for Local Company increased from 130 bps to 170 bps. Which of the following is the most likely action that the counterparties will request on their credit value adjustment (CVA)?

- A. The credit qualities of the counterparties have changed, but not enough to justify amending existing CVA arrangements.
- B. Global Bank requests an increase in the CVA charge it receives.
- C. Local Company requests a reduction in the CVA charge it pays.
- D. CVA is no longer a relevant factor, and the counterparties will use other mitigants of counterparty risk.

Q-61. An investor said: "The formula for BCVA is similar to the formula for CVA, except that the BCVA formula incorporates the probability of the counterparty's survival and uses EPE." Which of the following statements is accurate?

- A. The statement is correct with regard to EPE only.

- B. The statement is correct with regard to probability of survival only.
- C. The statement is incorrect with regard to both EPE and probability of survival.
- D. The statement is correct with regard to both EPE and probability of survival.

Q-62. ADB Banking Corporation (ADB) often enters into interest rate swaps with HIP Bank (HIP) on terms that reflect appropriate counterparty risk. Earlier in the year, HIP and ADB entered into a 3-year swap in which ADB agreed to pay HIP 5% fixed in return for 6-month LIBOR plus a spread. Since the swap was entered into, both banks were downgraded. As a result of the ratings changes, the credit spread for HIP has increased from 36 bps to 144 bps, while the credit spread for ADB has increased from 114 bps to 156 bps. Assuming no change in the LIBOR curve, if an identical 3-year swap was entered into today, which of the following is the most likely to be correct?

- A. Since HIP's spread increased more than ADB's spread, HIP's DVA will be higher and ADB's DVA will be lower.
- B. Since HIP's spread increased more than ADB's spread, HIP's CVA will be higher and ADB's CVA will be lower.
- C. Since both banks' spreads increased, the CVA on both sides of the contract will be higher.
- D. Since both banks' spreads increased, the DVA on both sides of the contract will be lower.

Q-63. A CRO at an investment bank has asked the risk department to evaluate the bank's 3-year derivative exposure position with a counterparty. The 1-year CDS on the counterparty is currently trading at a spread of 180 bps. The table below presents trade and forecast data on the CDS spread, the expected positive exposure, and the recovery rate on the counterparty:

	Year 1	Year 2	Year 3
Expected positive exposure (AUD million)	15	15	15
CDS spread (bps)	180	300	420
Recovery rate (%)	85	75	65

Additionally, the CRO has presented the risk team with the following set of assumptions to use in conducting the analysis:

- The counterparty's time-to-default follows a distribution of constant hazard rate.
- The investment bank and the counterparty have signed a credit support annex (CSA) to cover this exposure, which requires collateral posting of AUD 13 million throughout the life of the contract
- The current risk-free rate of interest is 2% and the term structure of interest rates remains flat over the 3-year horizon
- Collateral and exposure values will remain stable over the life of the contract.

Given the information and the assumptions above, what is the correct estimate for the credit valuation adjustment for this position?

- A. -AUD 0.140 million
- B. -AUD 0.172 million
- C. -AUD 0.442 million
- D. -AUD 1.051 million

Q-64. The following table illustrates how a financial institution might reconsider its stress test of current exposure in an expected-loss framework:

	PD	Stressed PD	Average EPE (millions)	LGD
Counterparty AA	0.05%	0.50%	\$213.00	70%
Counterparty BB	0.03%	0.30%	\$202.50	60%
Counterparty CC	0.45%	0.62%	\$75.00	70%
Counterparty DD	1.00%	1.60%	\$300.00	65%

What should be the Stress Loss value for Counterparty DD?

- A. \$420,000
- B. \$935,500
- C. \$1,170,000
- D. \$4,800,000

Q-65. A risk manager at Bank Gamma is presenting to a group of newly hired junior risk analysts on calculating bilateral CVA (BCVA). To illustrate the calculations, the manager assumes that Bank Gamma and Bank Phi are the only counterparties to each other and provides the following information about Bank Gamma:

- The discounted expected positive exposure to Bank Phi is CNY 60 million
- The discounted expected negative exposure to Bank Phi is -CNY 45 million

Additional information on the two banks is shown below:

Parameter	Bank Gamma	Bank Phi
Annual probability of default	2.5%	1.8%
Recovery rate	82%	92%

What is the BCVA from Bank Gamma's perspective?

- A. CNY 84,240
- B. CNY 114,615
- C. CNY 198,855
- D. CNY 201,960

Q-66. In credit risk management, Credit Valuation Adjustment (CVA) and Debit Valuation Adjustment (DVA) reflect adjustments to derivative valuations due to counterparty credit

risk and the institution's own credit risk, respectively. Under market stress, changes in credit spreads and exposures can impact these adjustments. During a market stress event:

- Both the financial institution's (FI) and its counterparty's (CP) credit spreads widen significantly.
- The FI's Expected Positive Exposure (EPE) to the CP increases.
- The FI's Expected Negative Exposure (ENE) (the FI's liability to the CP) remains unchanged.

How do the FI's CVA and DVA change under these conditions?

- A. CVA increases; DVA decreases.
- B. CVA increases; DVA increases.
- C. CVA increases; DVA remains unchanged.
- D. CVA decreases; DVA increases.

2.12. Key Point: Wrong-Way Risk

2.12.1. 重要知识点

2.12.1.1. Wrong-Way Risk: An outcome of any association, dependence, linkage, or interrelationship between exposure and counterparty creditworthiness that generates an overall increase in counterparty risk and, therefore, an increase in the amount of the credit value adjustment (CVA).

2.12.1.2. Right-Way Risk: Any dependence, linkage, or interrelationship between the exposure and default probability of a counterparty producing an overall decrease in counterparty risk is described as right-way risk.

2.12.2. 基础题

Q-67. Which of the following activities or transactions would most likely result in right-way risk with counterparty?

- A. Purchasing a put option from an A-rated company on that company's stock.
- B. Entering Into a forward contract to buy West Texas Intermediate (WTI) crude oil from an airline company at a fixed price
- C. Entering into a forward contract to buy WTI crude oil from a large oil producer at a fixed price.
- D. Selling a put option to an A-rated company on that company's stock.

Q-68. Bank HJK has written puts on Bank PQR stock to a hedge fund and sold CDS protection on Bank PQR to a manufacturer. Bank HJK and Bank PQR operate in several of the same businesses and geographies and their performances are highly correlated. Many in the

33-90

market are concerned that rising interest rates could negatively impact the credit quality of Bank HJK's numerous borrowers, which in turn would increase the credit spread of Bank HJK. From the perspectives of the hedge fund and the manufacturer, which of the following is correct with respect to their counterparty risk exposure to Bank HJK?

Hedge Fund	Manufacturer
A. Right-way risk	Wrong-way risk
B. Wrong-way risk	Right-way risk
C. Right-way risk	Right-way risk
D. Wrong-way risk	Wrong-way risk

Q-69. A hedge fund manages a portfolio of equity options. Among them are options written by a financial institution on its own stock. Assuming that all of the following options have the same expiration date and each of them corresponds to 1 share of the underlying stock of that financial institution, which of the long position in those options would give the highest wrong-way risk to the hedge fund?

- A. An in-the-money call option
- B. An in-the-money put option
- C. An out-of-the-money call option
- D. An out-of-the-money put option

Q-70. Which of the following is true about the impact of wrong-way risk (WWR) on central counterparties (CCPs)?

- A. Larger dealers represent right-way risk.
- B. CCPs are somewhat immune to wrong-way risk.
- C. CCPs are at risk of wrong-way risk (WWR) because they disassociate credit quality and Exposure.
- D. Their method of collateral acceptance promotes right-way risk with respect to posted Collateral.

2.13. Key Point: Mitigation of Counterparty Risk

2.13.1. 重要知识点

2.13.1.1. Mitigation of Counterparty Risk

- Netting Arrangement
- Collateralization
 - Threshold: an exposure level below which collateral is not called. It represents an amount of uncollateralized exposure.
 - Minimum transfer amount: the minimum quantity or block in which collateral may be

transferred. Quantities below this amount represent uncollateralized exposure.

- Independent amount: an amount posted independently of any subsequent collateralization. This is also referred to as the initial margin.

➤ Contract Clauses

- Walkaway
- Termination

➤ Central Counterparties

- Loss waterfall: Losses are first absorbed by the defaulted member's initial margin and default fund contributions. If losses are greater, the CCP equity and survivor members' default fund are used.
- Initial Margin: beginning deposit required from all CCP members to cover future potential default losses in a worst-case scenario; Variation margin: an additional margin for daily changes in asset positions.

2.13.2. 基础题

Q-71. A junior risk analyst at a consulting firm is reviewing the operational arrangements of bilateral netting and central clearing of derivative trades. The analyst examines the following bilateral trades of three firms:

- Firm 1's exposure to Firm 2: AUD 90 million;
- Firm 1's exposure to Firm 3: AUD 12 million;
- Firm 2's exposure to Firm 1: AUD 60 million;
- Firm 2's exposure to Firm 3: AUD 57 million;
- Firm 3's exposure to Firm 1: AUD 70 million;
- Firm 3's exposure to Firm 2: AUD 0 million;

Which of the following statements is correct?

- A. Under bilateral netting, Firm 1's net exposure is AUD 28 million.
- B. Under bilateral netting, Firm 2's net exposure is AUD 27 million.
- C. Under central clearing, Firm 3's net exposure is AUD 0 million.
- D. Under central clearing, the CCP's net exposure is AUD 28 million.

Q-72. A financial institution has many open derivative positions with an investment company. A description and current market values are displayed in the table below:

Position	Price (USD)
Long swaptions	32 million
Long credit default swaps	12 million
Short currency derivatives	16 million
Short interest rate swaps	8 million

In the event that the investment company defaults, what would be the loss to the financial institution if netting is used compared to the loss if netting is not used?

- A. Loss of USD 20 million if netting is used; loss of USD 24 million if netting is not used
- B. Loss of USD 20 million if netting is used; loss of USD 44 million if netting is not used
- C. Loss of USD 24 million if netting is used; loss of USD 32 million if netting is not used
- D. Loss of USD 24 million if netting is used; loss of USD 44 million if netting is not used

Q-73. The exhibit below presents a summary of bilateral mark-to-market (MtM) trades for four counterparties. If netting agreements exist between all pairs of counterparties shown, what is the correct order of net exposure per counterparty, from lowest to highest?

MtM Trades for Four Counterparties (USD million)				
		Opposing Counterparty		
		B	C	D
Counterparty A	Trades with positive MtM	10	10	1
	Trades with negative MtM	-10	-5	-10
		A	C	D
Counterparty B	Trades with positive MtM	10	0	10
	Trades with negative MtM	-10	-5	-2
		A	B	D
Counterparty C	Trades with positive MtM	5	5	2
	Trades with negative MtM	-10	0	-1
		A	B	C
Counterparty D	Trades with positive MtM	10	2	1
	Trades with negative MtM	-1	-10	-2

- A. A – C – B – D
- B. A – D – C – B
- C. D – B – C – A
- D. B – C – D – A

Q-74. Miven Corp. has two trades outstanding with one of its counterparties. Which of the following scenarios would result in the greatest netting advantage for Miven?

- A. The two trades have strong positive correlation.
- B. The two trades have weak positive correlation.
- C. The two trades are uncorrelated with each other.
- D. The two trades have strong negative correlation.

Q-75. A diversified portfolio of OTC derivatives with a single counterparty currently has a net mark-to-market value of USD 20,000,000 and a gross absolute mark-to-market value (the sum of the value of all positive-value positions minus the value of all negative-value

positions) of USD 80,000,000. Assuming there are no netting agreements in place with the counterparty, determine the current credit exposure to the counterparty.

- A. Less than or equal to USD 19,000,000
- B. Greater than USD 19,000,000 but less than or equal to USD 40,000,000
- C. Greater than USD 40,000,000 but less than USD 60,000,000
- D. Greater than USD 60,000,000

Q-76. What are the benefits of novation?

- A. Allows both party to walk away from the contract in case of default.
- B. A bilateral contract specifying that upon default, the non-defaulting party nets gains and losses with the defaulting counterparty to a single payment for all covered transactions
- C. Financial market contracts can be terminated upon an event of default prior to the bankruptcy process.
- D. Obligations are amalgamated with others

Q-77. You are the credit risk manager for a bank and are looking to mitigate counterparty credit risk exposure to ABCD, an A-rated firm. Currently your bank has the following derivatives contracts with ABCD:

Contract	Contract Value (HKD)
A	20,000,000
B	30,000,000
C	14,000,000
D	1,000,000

With the information provided, what is the most appropriate credit risk mitigation technique in this case?

- A. Implement a netting scheme
- B. Use credit triggers.
- C. Sell credit default swaps on ABCD
- D. Increase collateral.

Q-78. A financial institution has a two-way collateral support annex (CSA) with a counterparty covering a portfolio valued at JPY 400 million. The margining terms of the collateralized portfolio include a threshold of JPY 180 million, a minimum transfer amount of JPY 30 million, and a margin period of risk of 10 days. Which of the following is correct regarding the size of collateral in mitigating the counterparty risk of the portfolio?

- A. A lower threshold value is equivalent to a larger portion of exposure protected by collateral.

- B. A shorter margin period of risk is equivalent to a smaller portion of exposure protected by collateral.
- C. A lower independent amount is equivalent to a larger portion of exposure protected by collateral.
- D. The protection from collateral specified in the CSA is uniform throughout the life of the exposure profile.

Q-79. An underlying exposure with an effective annual price volatility of 6% is collateralized by a 10-year U.S. Treasury note with an effective price volatility of 8%. The correlation between the exposure and the U.S. Treasury note is zero. Changes in the value of the overall position (the exposure plus collateral) are calculated for a 10-day horizon at a 95% confidence interval. Which of the following would one expect to observe from this analysis?

- A. The presence of collateral increases the current exposure and increases the volatility of the exposure between remargining periods.
- B. The presence of collateral decreases the current exposure, but increases the volatility of the exposure between remargining periods.
- C. The presence of collateral increase the current exposure, but decreases the volatility of the exposure between remargining periods.
- D. The presence of collateral decreases the current exposure and decreases the volatility of the exposure between remargining periods.

Q-80. Rarecom is a specialist company that only trades derivatives on rare commodities. Rarecom and a handful of other firms, all of whom have large notional outstanding contracts with Rarecom, dominate the market for such derivatives. Rarecom management would like to mitigate its overall counterparty exposure, with the goal of reducing it to almost zero. Which of the following methods, if implemented, could best achieve this goal?

- A. Ensuring that sufficient collateral is posted by counterparties.
- B. Diversifying among counterparties.
- C. Cross-product netting on a single counterparty basis.
- D. Purchasing credit derivatives, such as credit default swaps.

Q-81. Consider the classic repurchase agreement (repo) in which Bank A is the repo seller who borrows cash and Counterparty B is the repo buyer who lends cash. Each of the following is true about this classic repo except which is false?

- A. A higher haircut reduces or mitigates the counterparty risk faced by the lender

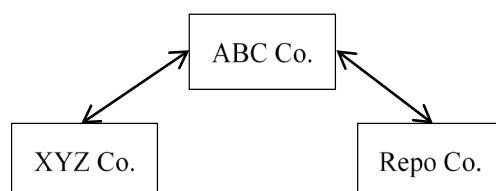
Counterparty B.

- B. The securities sold by Bank A become collateral legally owned by Counterparty B who, in the event Bank A's default, can liquidate the collateral in order to recover the cash lent.
- C. The repo seller Bank A borrows cash and therefore will not incur counterparty risk: only Counterparty B can incur counterparty risk.
- D. The repo buyer Counterparty B is exposed, at a minimum, to credit risk (default by Bank A), market risk (collateral price volatility), liquidity risk (collateral illiquidity), operational risk, and legal risk.

Q-82. XYZ, a small investment management firm, specializes in structuring small business loans and selling the government guaranteed portion to other institutional investors while retaining the riskier portions for high net worth investors. XYZ funds its operations by engaging in overnight repurchase agreements (repos) with three firms, but primarily with ABC, a firm that specializes in pooling funds from community banks and local government agencies and investing them in short-term, high-quality, government-secured investments.

Last week, XYZ was informed by ABC that its line had been frozen. XYZ learned that ABC had been defrauded by Repo Co., another repo borrower, who had provided false documentation of non-existent collateral of government-guaranteed loans. ABC feared a run by its investors as news of the fraud spread.

The diagram below illustrates the parties involved:



The use of a central clearinghouse to handle the transactions executed between XYZ's main funding source, ABC and ABC's client, Repo Co., would likely have resulted in a reduction in:

- A. ABC's funding liquidity risk.
- B. Repo Co.'s default risk.
- C. XYZ's lending risk.
- D. ABC's operational risk.

Q-83. By using a clearinghouse to handle the repo transactions between ABC and Repo Co., obligations owned between the two could have been netted once the fraudulent documentation was discovered. Which of the following is the most appropriate type of netting to use in this situation and what would be a likely additional impact from using

39-90

this netting?

- A. Payment netting would be used, which would reduce ABC's counterparty risk, but this risk would be transferred to other creditors outside the clearinghouse.
- B. Payment netting would be used, which would reduce Repo Co's counterparty risk, but ABC's counterparty risk would be increased.
- C. Closeout netting would be used, which would reduce ABC's counterparty risk, but this risk would be transferred to other creditors outside the clearinghouse.
- D. Closeout netting would be used, which would reduce Repo Co's counterparty risk, but ABC's counterparty risk would be increased.

Q-84. An investment bank has a one-way credit support annex (CSA) on a bilateral transaction with a hedge fund counterparty. Under the terms of the CSA, the mark-to-market value of the transaction forms the basis of the hedge fund's collateral requirements, which are provided below:

	Value (CNY)
Mark-to-market value of net exposure	25,000,000
Mark-to-market value of collateral posted	10,800,000
Threshold amount	14,000,000
Minimum transfer amount	2,500,000
Rounding amount	10,000

Assuming the net exposure increases to CNY 27,000,000 and the mark-to-market value of collateral posted has not changed, how much additional collateral will the hedge fund have to post?

- A. CNY 0
- B. CNY 1,990,000
- C. CNY 2,000,000
- D. CNY 2,500,000

Q-85. Given the following three events, what is the proper order of the CCP loss waterfall?

- I. Non-defaulting member's default fund contributions are exhausted.
- II. Defaulting member's collateral and default fund contributions are exhausted.
- III. CCP taps an amount of its equity that enables them to function normally.

- A. I, II, III.
- B. II, I, III.
- C. II, III, I.
- D. III, II, I.

Q-86. Following the financial crisis of 2007-2009, the roles of central counterparties (CCPs)

40-90

were increased to reduce systemic risk through a centralized clearing process. Which of the following actions is not an advantage of the CCP in the centralized clearing process?

- A. Increase transparency.
- B. Manage loss mutualization.
- C. Eliminate counterparty risk.
- D. Improve operational efficiency.

Q-87. What is a key difference between initial margin and default fund requirements for clearing members?

- A. Initial margin provides coverage with a high level of confidence, while default funds are used to mitigate extreme losses through mutualization, thus making clearing cost-effective but potentially increasing moral hazard.
- B. Initial margin requirements are significantly larger compared to default funds, which may lead to increased moral hazard but ensure better behavior, whereas default funds contribute to overall loss absorbency and systemic stability.
- C. Initial margin breaches are extremely rare and can lead to substantial losses which spreading risk among clearing members and clients.
- D. Default funds are updated more frequently than initial margins and are typically calibrated via stress tests to cover potential losses from multiple clearing member defaults and extreme market conditions.

Q-88. In the context of Central Counterparties (CCPs), which play a critical role in mitigating systemic risk in financial markets by acting as intermediaries between buyers and sellers, the loss absorption waterfall defines the sequence of financial resources used to cover losses from a member's default. This mechanism ensures market stability and protects non-defaulting members. Now a CCP with 10 clearing members faces a default by one member, resulting in a total loss of 75 units . Key parameters include:

- Each member's Initial Margin (IM) = 10.
- Each member's Default Fund (DF) contribution = 20.
- CCP's Skin-in-the-Game Equity (SIG) = 25.

What is the remaining Default Fund Reserve after absorbing the loss?

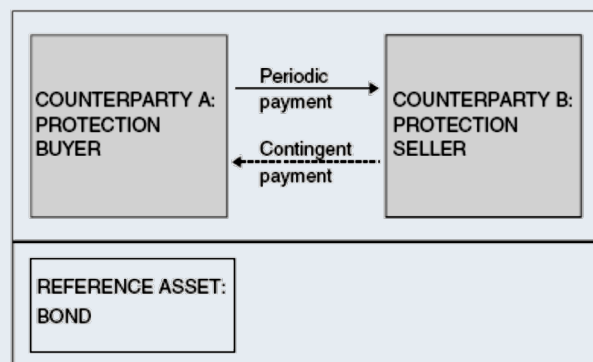
- A. 180
- B. 160
- C. 140
- D. 120

2.14. Key Point: Credit Derivatives 1 – CDS

2.14.1. 重要知识点

2.14.1.1. Credit Derivatives 1 – CDS

- In a Credit Default Swaps contract, a protection buyer (say A) pays a premium to the protection seller (say B), in exchange for payment if a credit event occurs.
- A default swap acts like a put option on the reference obligation for the buyer of the swap. If there is a default, the buyer receives a payment, which limits the buyer's downside risk.
- A long position in a corporate bond is equivalent to a long position in a risk-free bond plus a short position in a credit default swap.



- Settlement
 - Cash-Settled
 - Physically Settled
- CDS-bond basis

$$\text{CDS-Bond Basis} = \text{CDS Spread} - \text{Bond Yield Spread}$$

➤

2.14.2. 基础题

Q-89. When an institution has sold exposure to another institution (i.e., purchased protection) in a CDS, it has exchanged the risk of default on the underlying asset for which of the following?

- A. Default risk of the counterparty
- B. Default risk of a credit exposure identified by the counterparty
- C. Joint risk of default by the counterparty and of the credit exposure identified by the counterparty
- D. Joint risk of default by the counterparty and the underlying asset

Q-90. A six-year CDS on a AA-rated issuer is offered at 150bp with semiannual payments while the yield on a six-year annual coupon bond of this issuer is 8%. There is no counterparty risk on the CDS. The annualized LIBOR rate paid every six months is 4.6% for all maturities.

Which strategy would exploit the arbitrage opportunity? How much would your return exceed LIBOR?

- A. Buy the bond and the CDS with a risk-free gain of 1.9%.
- B. Buy the bond and the CDS with a risk-free gain of 0.32%.
- C. Short the bond and sell CDS protection with a risk-free gain of 4.97%.
- D. There is no arbitrage opportunity as any apparent risk-free profit is necessarily compensation for being exposed to the credit risk of the issuer.

Q-91. Which of the following statements concerning credit default swap is correct?

- A. The effective maturity of a CDS is reduced by including a mutual put in the master agreement.
- B. To purchase a single-name CDS contract, the buyer must also own the reference asset.
- C. Typically, there is only one reference asset that can be delivered to satisfy a CDS contract.
- D. CDS contracts can only be satisfied through cash settlement.

Q-92. The head of the structured securities desk at an investment bank directs all derivatives traders on the desk to use trade compressions on all eligible trades to reduce counterparty risk. One of the analysts decides to use trade compression on a portfolio of single-name CDS contracts with the same maturity and transacted with two counterparties to estimate the investment bank's compressed notional value:

Counterparty	Reference credit	Notional (EUR million)	Bank's position	Coupon (bps)
Digital Corporation	NUMU Inc.	9	Long protection	180
Digital Corporation	WKL Company	6	Long protection	150
WKL Company	NUMU Inc.	7	Short protection	150

What is the notional value of the investment bank's net contract on compressed trades only?

- A. EUR 1 million
- B. EUR 2 million
- C. EUR 8 million
- D. EUR 15 million

Q-93. Which of the following statements about CDS-bond basis is true?

- A. The CDS-bond basis refers to the difference between the credit default swap (CDS) spread and the bond yield spread, which ideally should be close to zero but is influenced by factors such as market pricing discrepancies, counterparty risk in CDS contracts, and

accrued interest exclusion.

- B. The CDS-bond basis represents the correlation between the credit default swap (CDS) spread and the bond yield spread, typically resulting in a negative value due to factors like market liquidity constraints and restructuring clauses in CDS contracts.
- C. The CDS-bond basis measures the convergence of credit default swap (CDS) spreads and bond yield spreads, with historical data showing a predominantly positive value before the Global Financial Crisis but mostly negative values post-crisis, influenced by factors such as counterparty risk and different reference rates.
- D. The CDS-bond basis denotes the average annual probability of default over a certain period, calculated using credit spreads and expected recovery rates, taking into account the relationship between spread, recovery rate, and hazard rate, which quantifies the likelihood of default within a given time frame.

Q-94. Which of the following statements accurately evaluates the use of credit indices and fixed coupons in pricing CDS transactions?

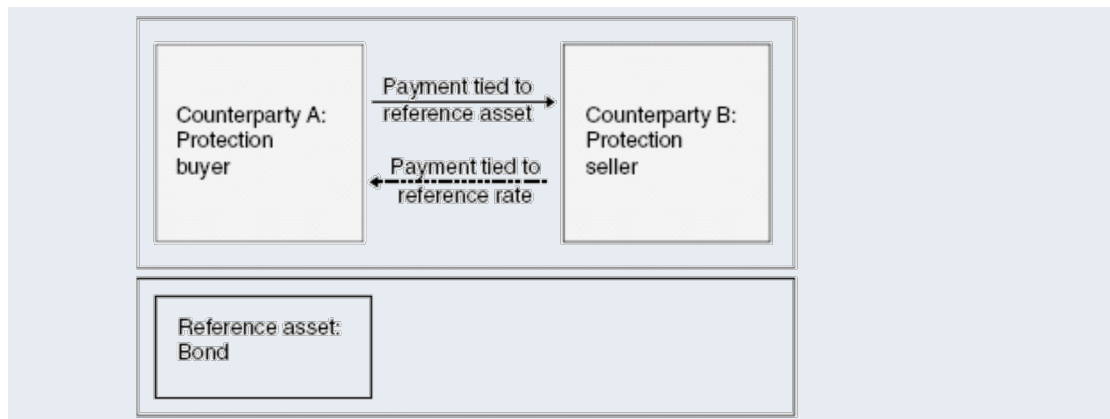
- A. Credit indices provide a precise average of CDS spreads for individual companies, while fixed coupons ensure consistent payments regardless of market fluctuations.
- B. Credit indices simplify trading by offering an average CDS spread for a portfolio of companies, and fixed coupons facilitate predictable payments throughout the contract duration.
- C. Credit indices consolidate CDS spreads for individual companies, ensuring uniformity in pricing, whereas fixed coupons adjust payments based on market conditions.
- D. Credit indices track individual CDS spreads for diverse companies, while fixed coupons stabilize payments by accounting for variations in market volatility.

2.15. Key Point: Credit Derivatives 2 – TRS

2.15.1. 重要知识点

2.15.1.1. Credit Derivatives 2 – TRS

- Total Rate of Return Swaps are contracts where one party, called the protection buyer (also called TROR payer and risk seller), makes a series of payments linked to the total return on a reference asset.



2.15.2. 基础题

- Q-95.** Risk Averse Bank (RAB) has made a loan of USD 100 million at 8% per annum. RAB wants to enter into a total return swap under which it will pay the interest on the loan plus the change in the mark-to-market value of the loan, and in exchange, RAB will get LIBOR + 30 basis points. Settlement payments are made annually. What is the cash flow for RAB on the first settlement date if the mark-to-market value of the loan falls by 2% and LIBOR is 6%?
- A. Net inflow of USD 0.3 million
 - B. Net outflow of USD 0.3 million
 - C. Net inflow of USD 1.7 million
 - D. Net outflow of USD 1.7 million
- Q-96.** A firm has entered into a USD 20 million total return swap on the NASDAQ 100 index as the index payer with ABC Corporation, which will pay 1-year LIBOR + 2.5%. The contract will last 1 year, and cash flows will be exchanged annually. Suppose the NASDAQ 100 Index is currently at 2,900 and LIBOR is 1.25%. The firm conducts a stress test on this total return swap using the following scenario:
- NASDAQ 100 in 1 year: 3,625
- LIBOR in 1 year: 0.50%
- For this scenario, what is the firm's net cash flow in year 1?
- A. A net cash outflow of USD 4.40 million.
 - B. A net cash outflow of USD 4.25 million.
 - C. A new cash inflow of USD 4.25 million.
 - D. A new cash inflow of USD 4.40 million.
- Q-97.** The credit protection buyer in a total return swap, where the protection seller pays LIBOR plus a spread, is hedged against each of the following risks except for:

- A. Credit deterioration in the reference.
- B. Spread risk in the reference.
- C. General level of interest rates.
- D. Counterparty exposure to the protection seller.

2.16. Key Point: Synthetic Collateralized Debt Obligation

2.16.1. 重要知识点

2.16.1.1. Synthetic CDO

- A synthetic collateralized debt obligation (synthetic CDO) is a complex financial instrument that mimics the characteristics of a traditional CDO but instead of holding actual underlying debt instruments, it is built on credit default swaps (CDSs) referencing a portfolio of credits, thereby transferring credit risk without direct ownership of the referenced assets.
- Spread Payments Approach and Gaussian Copula Model

2.16.2. 基础题

Q-98. Which of the following accurately describes the process of valuing a synthetic CDO using the spread payments approach and the Gaussian copula model?

- A. The Gaussian copula model is initially applied to calculate joint default probabilities among the reference entities, then these probabilities are used to determine the expected cash flows from the CDO tranches which, in turn, are discounted back to present value based on the spread payments approach.
- B. First, the spread payments approach calculates individual entity's default intensities, and then the Gaussian copula model synthesizes these into a multidimensional distribution to estimate the likelihood of simultaneous defaults; this information is used to price the CDO tranches.
- C. In this method, the spread payments approach initially estimates the fair spreads for each tranche based on historical data. Subsequently, the Gaussian copula model predicts the time to default for each referenced entity, and these predictions are used to adjust the estimated cash flows.
- D. The valuation process starts by utilizing the spread payments approach to establish an overall risk premium for the CDO. Then, the Gaussian copula model simulates correlated defaults across the portfolio to reallocate this premium among the different tranches according to their specific credit risks.

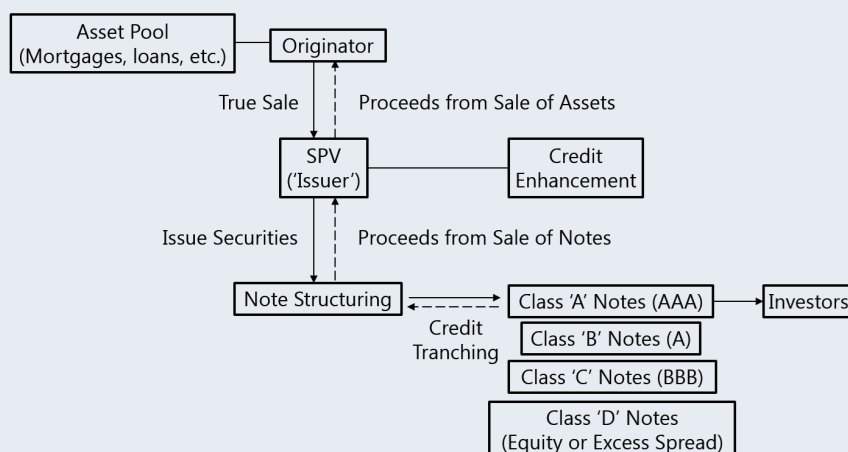
2.17. Key Point: Structured Products & CDO

2.17.1. 重要知识点

2.17.1.1. Structured Products & CDO

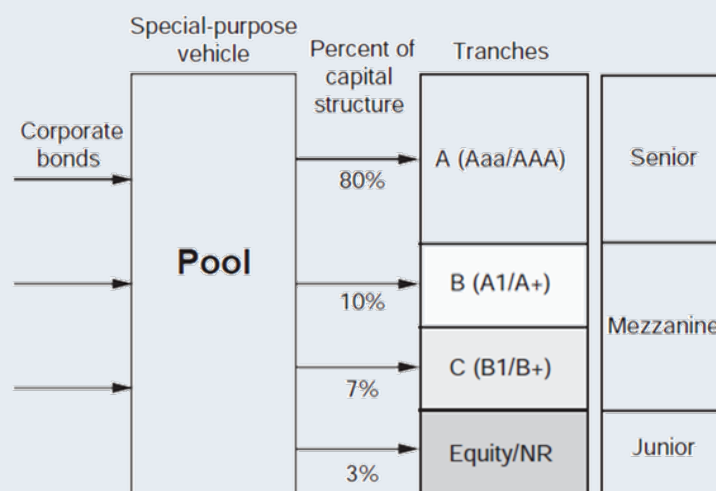
➤ Types of Structured Finance

● Securitization



➤ Structure

● Tranching



● Waterfall

The term “waterfall” is used because the capital structure is paid in a “top down” sequence with the senior debt receiving all of its promised payments before any lower tranche receives any monies.

➤ Structured Products

- Covered Bond
- MBS
- CDO

➤ Internal credit enhancement

- Subordination
- Over collateralization (O/C): more assets are pledged to back the structure and exceed the liabilities
- Excess spread: a positive excess spread between the collateral assets and the liabilities

(coupons) of the SPV, less fees and expenses.

➤ Effect of PD and Default Correlation

- The mezzanine effect is mixed.
- Constant Correlation: Increasing the probability of default will negatively impact the cash flows and, thus, the values of all tranches; Increasing default probability generally decreases the VaR for the equity tranches (less variation in returns) and increases the VaR for the senior tranches (more variation in returns).
- Constant Probability of Default: The equity tranches increases in value from increasing correlation as the possibility of zero (or few) credit losses increases from the high correlation. Senior VaR increases with correlation. As the default correlation approaches one, the equity VaR increases steadily. The interpretation is that although the mean return is increasing so is the risk as the returns are more variable (large losses or very small losses).

➤ Performance Analysis

Performance Measure	Calculation	Typical Asset Class
Public Securities Association (PSA)	$PSA = [CPR / (0.2)(months)] * 100$	mortgages, home-equity, student loans
Constant prepayment rate (CPR)	$1 - (1 - SMM)^{12}$	mortgages, home-equity, student loans
Single monthly mortality (SMM)	Prepayment / Outstanding pool balance	mortgages, home-equity, student loans
Weighted average life (WAL)	$\sum(a/365) * PF(s)$	mortgages
Weighted average maturity (WAM)	Weighted maturity of the pool	mortgages
Weighted average coupon (WAC)	Weighted coupon of the pool	mortgages
Debt service coverage ratio (DSCR)	Net operating income / Debt payments	commercial Mortgages
Monthly payment rate (MPR)	Collections / Outstanding pool balance	all non-amortizing asset classes
Default ratio	Defaults / Outstanding pool balance	credit cards
Delinquency ratio	Delinquents / Outstanding pool balance	credit cards
Absolute prepayment speed (ABS)	Prepayments / Outstanding pool balance	auto loans, truck loans
Loss curves	Show expected cumulative loss	auto loans, truck loans

2.17.2. 基础题

- Q-99.** A bank is planning to securitize car loans by creating an SPV. The bank would sell the loans to an SPV through a “true sale” and the SPV would issue securities under a “revolving securitization structure.” The project manager is reviewing characteristics of securitization transactions in general as well as specific features that are commonly incorporated into revolving structures. Which of the following statements is correct?
- A. The credit quality of the securitized car loan assets would be enhanced if the principal value of securities issued is higher than the principal value of the assets.
 - B. The SPV can allow the bank to access cheaper funding if the credit quality of the securitized car loan assets is higher than the credit quality of the bank’s balance sheet.

- C. Under a revolving structure, prepayment assumptions are not incorporated, which typically results in principal amounts paid to investors through a series of coupons over the lifetime of the security.
- D. Under a revolving structure, the bank transfers the car loan assets to the SPV and the SPV can issue multiple securitizations, which are priced and traded based on weighted-average life of the structure.

Q-100. An endowment fund has sold default protection on the most senior tranche of a CDO. If the default correlation between assets held in the CDO decreases sharply from the correlation used in pricing the CDO tranches, assuming everything else is unchanged, how will the position of the endowment fund be impacted?

- A. It will either increase or decrease, depending on the pricing model used and the market conditions.
- B. It will gain significant value, since the probability of exercising the protection falls.
- C. It will lose significant value, since the protection will gain value.
- D. It will neither gain nor lose value, since only expected default losses matter and correlation does not affect expected default losses.

Q-101. National united bank has recently increased the bank's liquidity through securitization of existing credit card receivables. The proposed securitization includes tranches with multiple internal credit enhancements as shown in Exhibit 1 below. The total value of the collateral for the structure is USD 600 million, no lockout period, and the subordinated tranche B bond is the first loss piece:

Bond Class	par value
Senior tranche	USD 250 million
Junior tranche A	USD 200 million
Junior tranche B	USD 70 million
Subordinated tranche A	USD 50 million
Subordinated tranche B	USD 30 million
Total	USD 600 million

At the end of the fourteenth month after the securities were issued, the underlying credit card accounts have prepaid USD 300 million in principal in addition to regularly scheduled principal and interest payments. What is the amount of the prepaid principal paid out to the holders of the junior tranche A bond class?

- A. USD 0 million
- B. USD 50 million
- C. USD 120 million

D. USD 230 million

Q-102. King Motors Acceptance Corporation (KMAC), the finance arm of King Motors, issues an auto-loan asset-backed security that consists of a senior tranche, denoted Tranche A in the amount of \$50 million and an interest payment of 5 percent, and two subordinated tranches, denoted Tranches X and Z respectively, each with a face amount of \$35 million. Tranche X pays investors annual interest at a rate of 6.5 percent while Tranche Z pays investors annual interest at a rate of 7.5 percent. Which of the following methods of credit support would NOT affect the credit quality of subordinated Tranche X?

- A. The total amount of the auto loans that make up the asset-backed issue is \$125 million.
- B. The weighted average interest rate on the auto loans making up the pool is 6.4 percent.
- C. Any defaults on the part of King Motor's customers will be first absorbed by Tranche Z.
- D. KMAC has a reserve in the amount of \$10 million that will remain on KMAC's balance sheet.

Q-103. Assume the originator securitizes a \$100 million loan portfolio that pays LIBOR plus 200 bps. Senior expenses of the SPE amount to 20 bps. The SPE issues only two classes of securities: senior debt with face value of \$90 million and subordinated debt with face value of \$10 million, such that the subordinated debt "functions as equity". The coupon on the senior debt is LIBOR plus 100 bps. The subordinated debt (equity) gets an interest rate equal to the realized net excess spread. What is the net excess spread?

- A. \$10 million $\times$ (LIBOR + 3%)
- B. \$10 million $\times$ (LIBOR + 5%)
- C. \$10 million $\times$ (LIBOR + 7%)
- D. \$10 million $\times$ (LIBOR + 9%)

Q-104. In which of the below the assets remain on the balance sheet of the institution

- A. CMO
- B. CLO
- C. MBS
- D. Covered Bond

Q-105. Each of the following is a valid difference between a covered bond and a true securitization except which is not true?

- A. In a covered bond, the cover pool remains on the balance sheet, but in a true securitization, loans (assets) are removed from the balance sheet.
- B. In a covered bond, principal and interest are paid from issuer's general cash flows, but in

50-90

a securitization, principal and interest are paid from the collateral pool directly.

- C. Unlike a true securitization, there is not a “true sale” of assets to a bankruptcy-remote special purpose vehicle in the case of covered bond.
- D. Unlike a true securitization, a covered bond neither create securities nor is a genuine method for raising funds (i.e., borrowing) in capital markets.

Q-106. Which of the following statements about portfolio losses and default correlation are most likely correct?

- I. Increasing default correlation decreases senior tranche values but increases equity tranche values.
 - II. At high default rates, increasing default correlation decreases mezzanine bond prices.
- A. I only
 - B. II only
 - C. Both I and II
 - D. Neither I nor II

Q-107. A collateralized mortgage obligation (CMO) has the characteristics below. Which of the following are most accurate regarding its credit enhancement?

Return on assets	8.75%
Senior tranche	\$400,000,000
Subordinated tranche A	\$120,000,000
Subordinated tranche B	\$50,000,000
Value of collateral	\$600,000,000
Interest paid on liabilities of SPE	7.50%
Fees and expenses	0.60%

- I. There is overcollateralization.
 - II. The investors gain credit enhancement through the excess spread.
- A. I only.
 - B. II only.
 - C. Both I and II.
 - D. Neither I nor II.

Q-108. Which of the following is an internal enhancement?

- A. Overcollateralization
- B. CDS
- C. Put options on assets
- D. Letters of credit

- A. Residential mortgage.
- B. Credit card debt.
- C. Commercial mortgage.
- D. Commercial paper.

Q-112. A fixed-income portfolio manager purchases a seasoned 5.5% agency mortgage-backed security with a weighted average loan age of 60 months. The current balance on the loans is USD 20 million, and the conditional prepayment rate is assumed to be constant at 0.4% per year. Which of the following is closest to the expected principal prepayment this month?

- A. USD 1,000
- B. USD 7,000
- C. USD 10,000
- D. USD 70,000

Q-113. Assume an MBS is composed of the following four different pools of mortgages:

- \$2 million of mortgages that have a maturity of 90 days.
- \$3 million of mortgages that have a maturity of 180 days.
- \$5 million of mortgages that have a maturity of 270 days.
- \$10 million of mortgages that have a maturity of 360 days.

What is the weighted average maturity (WAM) of these mortgage pools?

- A. 167 days.
- B. 225 days.
- C. 252 days.
- D. 284 days.

Q-114. A manager from the structured credit risk desk at a bank is presenting to a group of newly hired risk analysts on calculating cash flows in a securitization structure. The manager illustrates the procedure with the existing collateral pool of loans and the corresponding liabilities, all with a maturity of 5 years, using the information in the table.

Initial number of loans in the collateral pool	100
Principal amount of each loan	EUR 1,000,000
Total coupon interest to be paid annually on all junior and senior bonds	EUR 6,300,000
Maximum annual amount flowing from the excess spread into the overcollateralization account	EUR 1,500,000
Swap rate per year for all maturities	3.5%
Recovery rate in the event of a loan default	45%

The manager makes additional observations as follows:

- The loans in the collateral pool pay a fixed spread of 2.2% over the swap curve
- There were no defaults in year 1
- The value of the overcollateralization account at the end of year 1 was EUR 0

What is the value of the overcollateralization account at the end of year 2 if there are 8 defaults in year 2?

- A. EUR 600,000
- B. EUR 1,056,000
- C. EUR 2,544,000
- D. EUR 3,600,000

Q-115. In a securitization structure, the Overcollateralization (OC) Account is designed to absorb losses and protect senior tranche investors. Excess interest cash flows (after paying senior and mezzanine tranches) are directed to the OC account, subject to an annual maximum inflow cap. Recoveries from defaulted loans are added to the OC account without being capped at the end of the year. Given:

- Asset Pool: 80 loans, each 625, 000 (total pool = 50 million).
- SOFR = 5.25%
- Loan Interest Rate: SOFR + 285 bps (total rate = 8.1%).
- Tranches:
 - ✓ Senior Tranche: \$32 million, pays SOFR + 85 bps (6.1%).
 - ✓ Mezzanine Tranche: \$11 million, pays SOFR + 375 bps (9%).
 - ✓ Equity Tranche: \$7 million.
- Defaults:
 - ✓ 2023: 0 defaults.
 - ✓ 2024: 6 defaults (total defaulted principal = $6 \times 625, 000 = 3.75$ million).
- Recovery Rate: 46% for both years (recovered amounts added to OC at year-end).
- OC Account Rules:
 - ✓ 2023 maximum inflow from excess interest: \$0.5 million.
 - ✓ 2024 maximum inflow from excess interest: \$0.75 million.
- Initial OC balance (2022): \$0.

Calculate the OC Account balance at the end of 2024 assuming it is reinvested at the end of each year with SOFR.

- A. 1.725 million
- B. 2.975 million
- C. 3.001 million
- D. 3.725 million

2.18. Key Point: Retail Credit Risk

2.18.1. 重要知识点

4.1.1.1. Credit Decision and Credit Analysis

➤ Key Variables in Mortgage Credit Assessment:

- FICO: Number score of the default risk associated with a borrower's credit history.
- DTI: Debt-to-income ratio is used to qualify mortgage payment and other monthly debt payments versus income.
- LTV: Expresses the amount of a first mortgage lien as a percentage of the total appraised value of the property – i.e., the loan-to-value ratio.
- Scorecard Performance:
- When measuring a scorecard's performance, the validation technique traditionally employed is the cumulative accuracy profile (CAP) and its summary statistic, the accuracy ratio (AR):

$$AR = \frac{A_R}{A_P}$$

2.18.2. 基础题

Q-116. In assessing the key variables associated with a potential mortgage loan, a bank will charge a higher interest rate if the borrower has a relatively:

- A. High FICO score.
- B. High loan-to-value ratio.
- C. Low debt-to-assets ratio.
- D. Low debt-to-income ratio.

Q-117. A risk manager of ABC bank is looking at the mechanics of a refinancing event. A house was originally purchased for USD 500,000 and financed with an interest-only hybrid adjustable rate mortgage (with a LTV of 80%) at an initial annual fixed rate of 2.75%. Before the interest-only period elapses, the house's appraised value drops to USD 400,000. The homeowner decides to refinance the existing mortgage with a 15-year fixed mortgage at an annual rate of 4.5% (LTV 80%) and uses cash to pay off the remainder of the original mortgage's principal balance. Assuming standard payments, what is the approximate change in monthly payments as a result of the refinancing?

- A. USD 1,500
- B. USD 1,900
- C. USD 2,400
- D. USD 3,000

Q-118. About the cumulative accuracy profile (CAP), each of the following statements is true except which is not?

- A. A perfect credit scoring model generates an accuracy ratio (AR) of 1.0, which is the upper bound on the AR.
- B. A purely random model that cannot differentiate between good and bad customers is likely to generate an accuracy ratio (AR) of 0.40 to 0.60; i.e., 50% +/- 10%.
- C. The CAP curve, which plots the actual rating model as a cumulative percentage of defaults, is monotonically increasing (aka, nondecreasing or weakly increasing).
- D. The CAP curve plots the fraction of defaulted customers (y axis) against the fraction of entire customer population sorted by score from highest risk (left) to lowest risk (right).

Q-119. A credit manager at a US-based commercial bank asks a team of risk analysts to examine the risks attributed to a large retail credit portfolio of the bank. The manager instructs the analyst to suggest measures to mitigate the “dark side” of retail credit risk affecting the portfolio. Which of the following would most likely be an effective measure?

- A. Focus the extension of credit on low default portfolios such as mortgages or large corporations.
- B. Concentrate on expected loss estimation since systematic risk factors such as a real estate crisis or a sharp economic downturn can be diversified away.
- C. Monitor the effectiveness of credit risk assessment tools for retail customers and adjust the tools as needed.
- D. Use stress tests to analyze the exposure to idiosyncratic risk factors of every single retail credit customer.

2.19. Key Point: Country Risk

2.19.1. 重要知识点

4.1.1.2. Country Risk

- Foreign Currency Defaults and Local Currency Defaults
 - Foreign currency defaults and local currency defaults have different implications, with local currency defaults often tied to a country's ability to print more currency.
 - Reasons for local currency default include constraints on printing currency, such as under the gold standard or shared currency arrangements like the Euro. Countries may face a trade-off between default and currency debasement, with implications for inflation and economic stability.
- Consequences of a Country's Default
 - Reputation loss, economic recessions, political instability, increased probability of banking crises, currency devaluations, and changes in political leadership. Default can

lead to long-lasting impacts on firms' investment capacity and institutional stability.

➤ Methods for Predicting Sovereign Default

- Credit ratings, market-based credit default spreads, and CDS spreads are used to measure country risk. Credit ratings provide a qualitative assessment, while market-based spreads offer real-time information but can be volatile. CDS spreads reflect market sentiment but can be influenced by liquidity.

2.19.2. 基础题

Q-120. In comparing foreign currency defaults and local currency defaults, which of the following statements is correct?

- A. Foreign currency defaults typically occur due to constraints on printing currency, while local currency defaults happen when countries lack the ability to print more currency.
- B. Foreign currency defaults involve debt denominated in the country's own currency, while local currency defaults occur when there are shortages of the foreign currency to meet obligations.
- C. Local currency defaults can occur even when countries can print more local currency, while foreign currency defaults are primarily driven by deliberate trade-offs between default and currency debasement.
- D. Local currency defaults primarily occur due to foreign currency shortages, while foreign currency defaults may happen despite the ability to print more currency.

Q-121. Which of the following statements accurately compares the use of credit ratings, market-based credit default spreads, and CDS spreads in predicting default?

- A. Credit ratings are the most reliable predictor of default as they provide a qualitative assessment of risk that is widely recognized and regularly updated.
- B. Market-based credit default spreads offer the dynamic and granular insight into default risk, though they may be influenced by factors unrelated to default risk such as liquidity and investor demand.
- C. CDS spreads are the preferred choice for predicting default due to their real-time updates and reflection of current information on default risk, despite their susceptibility to counterparty and liquidity risks.
- D. Market-based credit default spreads are superior to both credit ratings and CDS spreads in predicting default, as they offer real-time updates and granular risk differentiation without being influenced by factors unrelated to default risk.

Q-122. What are some of the shortcomings of the sovereign rating systems of rating agencies?

- A. Sovereign ratings are often too pessimistic, leading to unnecessary market panic.

- B. Ratings agencies tend to act independently, resulting in conflicting assessments of sovereign risk.
- C. Ratings agencies may delay or inadequately update sovereign ratings, leaving investors vulnerable to crises.
- D. Sovereign ratings are heavily influenced by governments, leading to biased assessments.

2.20. Key Point: Governance Practices

2.20.1. 重要知识点

4.1.1.3. Key Principles of Credit Risk Governance

- The key principles underlying best practice for the governance system of credit risk include guidelines, skills, limits, and oversight.
 - Guidelines refer to clear rules governing the approval of transactions involving credit risk.
 - Skills involve delegating authority to individuals or committees with appropriate expertise.
 - Limits represent the maximum risk exposure a company is willing to accept.
 - Oversight ensures that qualified staff with adequate independence and resources monitor and assess risk management activities.

4.1.1.4. Roles of the Credit Committee

- Decision-Making
 - The credit committee is responsible for making decisions on approving or rejecting transactions based on their credit risk assessment. They ensure that all aspects of a transaction are thoroughly evaluated before approval.
- Representation
 - The committee comprises individuals from various disciplines, including the business unit, risk management, legal, compliance, and accounting. This diverse representation ensures that all perspectives are considered in the decision-making process.
- Charter
 - The credit committee operates based on a charter that outlines its responsibilities, decision-making processes, and approval guidelines. This charter is typically part of the organization's credit risk management framework.
- Preparation
 - Originators are required to prepare an approval package for presentation to the credit committee. This package should cover all transaction aspects and be distributed well in advance to allow committee members time for review and inquiries.
- Chairperson
 - The credit committee is led by a respected chairperson who facilitates discussions,

ensures all parties have the opportunity to present their opinions, and maintains a neutral and objective environment during deliberations.

➤ Record-Keeping

- Detailed and accurate records of credit committee discussions are essential. Meeting minutes should be recorded and distributed promptly after each meeting to document decisions and ensure transparency.

2.20.2. 基础题

Q-123. In the governance system of credit risk management, what is the primary purpose of setting clear guidelines?

- A. Guidelines serve as punitive measures for breaches of credit risk management rules.
- B. Guidelines ensure that transactions involving credit risk are approved swiftly without thorough examination.
- C. Guidelines establish concise, accessible rules to facilitate compliance with credit risk management standards.
- D. Guidelines are complex documents which may confuse individuals involved in credit risk management.

Q-124. What is one of the key roles of the credit committee in an organization?

- A. The credit committee primarily focuses on financial aspects when evaluating transactions.
- B. The credit committee is responsible for evaluating transactions without considering diverse perspectives.
- C. The credit committee ensures that all aspects of transactions are thoroughly evaluated before approval.
- D. The credit committee is chaired by a member of the business unit to ensure biased decision-making.

Q-125. ABC Bank had a credit committee meeting yesterday, and the following is a copy of the notes taken during the meeting:

- All members are present Sarah (Chief Risk Officer), Edward (Head of Compliance), Jennifer (Head of Legal), David (Head of Tax), and Mark (Chief Financial Officer).
- Sarah initiated the meeting by confirming if everyone had received an email containing approval documents from credit officer Jack, who had originated the loan. Everyone confirmed receiving the email with the documents yesterday.
- Sarah provided an overview of the loan for approximately five minutes, after which she proceeded to inquire about the opinions of those present. It was the consensus of all individuals, except for Jennifer, that the loan be approved. Jennifer expressed

59-90

concern regarding the trustworthiness of the counterparty.

- Sarah respects Jennifer's opinion but disagrees and initiates a vote. Two other committee members agree with Sarah, resulting in loan approval.

What is the most probable issue with the credit committee's meeting?

- A. The approval documents should have been distributed earlier.
- B. Sarah monopolized the conversation and did not properly facilitate discussion.
- C. The meeting was not properly documented.
- D. The vote was not unanimous, so the loan should have been rejected.

2.21. Key Point: Loan and Financing Policies

2.21.1. 重要知识点

4.1.1.5. Effective Lending or Financing Policy

- Clear lending authority and limits for lending officers.
- Specific types of loans offered and guidelines for each.
- Formal appraisal procedures and requirements.
- Adequate loan pricing to cover costs and potential losses.
- Monitoring and reporting procedures for collections.
- Guidelines for sectoral analysis of loans.
- Recognition and management of loan impairments.
- Restrictions on total outstanding loans and loan-to-collateral ratios.
- Written internal guidelines for departments.
- Regular loan portfolio reviews to assess risk and performance.

4.1.1.6. Loan Loss Provisions and Loan Loss Reserves

- Loan loss provisions are amounts set aside by financial institutions to cover potential losses that may arise from loans that may default in the future.
- Loan loss reserves are a part of a bank's capital that is specifically set aside to cover potential future losses.

4.1.1.7. IFRS 9 Requirement

- When estimating expected credit losses, banks should consider the full lifetime expected credit losses for financial instruments that have experienced a significant increase in credit risk since initial recognition. For other financial instruments, banks should estimate expected credit losses over a 12-month period.

4.1.1.8. Retaining Loss Assets and Writing Off Loss Assets

- Retaining loss assets allows for ongoing recovery efforts through negotiation or legal actions, potentially improving long-term financial position, but it requires continuous resources and may impact liquidity and profitability in the short term.
- Writing off loss assets enhances financial reporting transparency by recognizing

irrecoverable losses and cleaning up the balance sheet, but it directly affects profitability and capital adequacy ratios, potentially overlooking future recovery possibilities.

2.21.2. 基础题

Q-126. Sarah is a credit officer at ABC Bank and is trying to get a \$135M loan approved for an R3 internally rated company. ABC Bank has the below approval hierarchy.

Internal Rating of Counterparty	Exposure in USD		
	R1	135	160
	R2	110	135
	R3	85	110
	R4	60	85
	R5	35	60
	R6	10	35
	Heading of Trading	Head of Risk	Credit Committee

Who will need to approve this loan?

- A. Head of Trading
- B. Head of Risk
- C. Credit Committee
- D. Head of Trading, Head of Risk, Credit Committee

Q-127. XYZ and ABC Bank each make a loan of \$1,000,000 at 4% interest to a technology startup, Innovate, in January of 2022. Unfortunately, Innovate makes only 4 payments. Innovate defaulted and became unlikely to repay its debt. ABC decides to retain the asset in hopes of recovering part of the loan one day while XYZ decides to write off the lost asset. Assume all other items between the banks are the same. James, a financial analyst, is evaluating both banks' financials. He wants to make the firm's financial statements comparable. What would he need to change if he believes that the loan to Innovate will never be recovered?

- A. Reduce ABC's asset values only.
- B. Increase ABC's loan loss reserves only.
- C. Reduce ABC's asset values and its loan loss reserves.
- D. Reduce ABC's asset values and increase its loan loss reserves.

Q-128. What is one of the challenges in setting exposure limits and concentration thresholds, as discussed in regulatory policies to limit credit risk?

- A. The difficulty in defining exposure does not affect the inclusion of contingent liabilities within the exposure limit.
- B. Establishing limits for related parties is typically stipulated by regulators, and banks are

required to maintain such limits as a matter of board policy, ensuring prudent banking practice and regulatory compliance.

- C. Modern prudential regulations usually mandate that banks refrain from investing in or extending credit to any individual entity or related group of entities in excess of a prescribed percentage of the bank's expected loss.
- D. The treatment of collateral is uncontroversial because the valuation of collateral is typically objective and straightforward.

Q-129. Which of the following statements about the scope and allocation of a bank's credit facilities and the management of its credit portfolio is correct?

- A. In larger banks, lending authority is usually centralized, whereas in smaller banks, it tends to be decentralized based on geographical area, lending products, and customer types.
- B. The lending policy should not establish limits for lending officers, allowing them flexibility in approving loans according to their authority and experience.
- C. Prudent banking practice dictates that facilities extended to related parties should not require board approval.
- D. A lending policy should provide an outline of the scope and allocation of credit facilities and the management of the credit portfolio, including the origination, appraisal, approval, disbursement, monitoring, collection, and handling procedures for various credit functions.

Solutions

Q-1. Solution: D

Option A is incorrect because it inaccurately states that insolvency occurs due to the failure to fulfill contractual obligations, which is the definition of default. Furthermore, it incorrectly suggests that bankruptcy aims to restructure debts and assets, whereas bankruptcy may also result in liquidation. Option B is incorrect because it reverses the definition of insolvency and inaccurately describes the outcome of default as leading to court intervention for dissolution.

Option C is incorrect because it conflates the concepts of default and bankruptcy and incorrectly suggests that bankruptcy primarily involves the liquidation of assets due to temporary illiquidity.

Option D correctly summarizes the distinctions among insolvency, default, and bankruptcy. It distinguishes each term accurately, reflecting their respective definitions and relationships.

Q-2. Solution: A

Option A accurately describes the modern nature of credit risk exposure in financial transactions. It highlights how derivatives and similar instruments can generate substantial credit exposure without immediate cash involvement, emphasizing the importance of risk analysis and hedging strategies. Options B, C, and D contain inaccuracies or misrepresentations of credit risk exposure. Option B incorrectly suggests that credit risk is primarily associated with traditional lending practices, while Option C misrepresents the level of credit risk in leasing transactions. Option D inaccurately states that deposits in banks are entirely risk-free, disregarding potential credit risk exposure beyond guaranteed insurance limits. Therefore, Option A is the correct choice.

Q-3. Solution: D

Reinsurance is essentially insurance for insurance companies. This introduces more credit risk for ABC (i.e., by transferring credit risk to a third party) because ABC will pay its premiums every month but is unsure if the reinsurance company will fulfill its payment obligations.

In regard to (A), (B), and (C), each is true; i.e., the statements are plausible.

Insurance companies have considerable amounts of money to invest, and every dollar invested has the possibility of not being paid back. Emerging market bonds have the possibility of default and, therefore, bear credit risk. Insurance companies originate policies and thus carry the risk of having to payout on those policies; but more specifically, David's particular concern about originating new policies "when we extend installment terms and grace periods to our corporate customers" is a reference to the company's apparent policy to generate accounts receivable (clearly a credit risk).

Q-4. Solution: A

Q-5. Solution: D

Default correlation has no effect on portfolio EL, but is an increasing function with portfolio UL.

Q-6. Solution: D

Q-7. Solution: B

The expected loss is \$20,000 (\$1,000,000 × 0.02). If there are three defaults, the credit loss is \$60,000 (3 × \$20,000). The credit VaR at the 95% confidence level is \$40,000 (calculated by taking the credit loss of \$60,000 and subtracting the expected loss of \$20,000).

Q-8. Solution: A

The one-month PD = $1 - (100\% - 4\%)^{(1/12)} = 0.3396\%$.

Expected Loss

$$= 98.9846\% \times 0 + 1.10119\% \times \$1.0 \text{ m} + 0.0034\% \times \$2.0 \text{ m} + 0\% \times \$3.0 \text{ m} = \$10,188$$

The probability of zero defaults = $(1 - 0.3396\%)^3 = 98.98464\%$.

Therefore, the 99.0% WCL is one default or \$1.0 million, and the 99.0% CVaR = \$1.0 million - \$10,188 = \$989,812.

Q-9. Solution: A

The 95% credit VaR corresponds to the unexpected loss at the 95th percentile minus the expected loss, or the expected future value at the 95% loss percentile minus the current value. Using the probabilities in the given ratings transition matrix, the 95% percentile corresponds to a downgrade to BBB, at which the value of the bond would be estimated at 101. Since cash flows for the bond are not provided, we cannot derive the precise expected and unexpected losses, but the credit VaR (the difference) is easily derived by subtracting the estimated value given a BBB rating from the current value. 95% credit VaR = 110 - 101 = 9.

Q-10. Solution: B

The 95th percentile of the credit loss distribution is \$28,000 ($28 \times \$1,000,000/1,000$). The expected loss is \$20,000 (\$1,000,000 × 0.02). The credit VaR is then \$8,000 (\$28,000 - \$20,000).

Q-11. Solution: C

$$\text{Default Correlation} = \frac{0.4\% - 2\% \times 3\%}{\sqrt{2\% \times 98\% \times 3\% \times 97\%}} = 0.142365$$

Q-12. Solution: A

A is correct. Since the 95th percentile of the WCL of the three loans are the same and the 95% CVaR is derived as follows:

95% CVaR = 95th percentile of the WCL – Expected Loss. Therefore, we only need to consider the expected loss.

The PD for each loan:

Loan S = Investment grade, 2-year maturity: PD = 1.5%

Loan T = Non-investment grade, 3-year maturity: PD = 12.0%

Loan U = Investment grade, 4-year maturity: PD = 3.5%

The expected loss (EL) :

Loan S: $EL = 0.015 \times 0.80 \times \text{SGD } 55,000,000 = \text{SGD } 660,000$

Loan T: $EL = 0.12 \times 0.90 \times \text{SGD } 36,000,000 = \text{SGD } 3,888,000$

Loan U: $EL = 0.035 \times 0.70 \times \text{SGD } 50,000,000 = \text{SGD } 1,225,000$

Since $CVaR = 95\text{th percentile loss} - EL$, we can conclude that: $CVaR \text{ of Loan S} > CVaR \text{ of Loan U} > CVaR \text{ of Loan T}$

Q-13. Solution: B

Option B accurately describes the Credit Risk Plus approach, which focuses solely on estimating default losses by assuming default probabilities of individual firms without explicitly modeling default correlation. Options A and C describe different aspects of Vasicek's Model and Credit Metrics, respectively, but they do not emphasize estimating default losses solely based on individual firm default probabilities without explicitly considering default correlation. About option D, the extension by Gordy is not based on the single-factor world assumption of the Vasicek model. Instead, Gordy's extension primarily relies on the Worst Case Default Rate (WCDR) for individual loans within a large portfolio, using relevant parameters to estimate high percentiles of the default rate distribution. This is done to calculate regulatory capital requirements more accurately, rather than relying on a single-factor assumption.

Q-14. Solution: C

$$UL = 50 \times \sqrt{0.02 \times 0.4^2 + 0.5^2 \times 0.02 \times 0.98} = 4.5$$

Q-15. Solution: B

$$\sqrt{5.5^2 + 5.5^2 + 2 \times 0.28 \times 5.5 \times 5.5} = 8.8$$

$$ULC = \frac{UL_1(UL_1 + \rho UL_2)}{UL_P} = 5.5 \times \frac{5.5 + 0.28 \times 5.5}{8.8} = 4.4$$

Q-16. Solution: A

A is correct. If the confidence level is raised, economic capital needed would be higher since a higher capital multiplier is applied to the UL.

B is incorrect. A lower confidence level implies a lower credit rating target (and a lower capital multiplier), and hence lower capital level is needed to cover credit assets, for both banks.

C is incorrect. See explanation in B.

D is incorrect. Given the same exposures and EL, and since asset correlation (hence, the volatility of UL) for Bank XYZ credit assets is higher, economic capital needed for Bank XYZ will be higher than that for Bank ABC even if the confidence level is unchanged (i.e., the capital multiplier is unchanged).

Q-17. Solution: B

A) This option incorrectly states that real-world probabilities are derived from theoretical models, which is not the case. Real-world probabilities are based on historical data and market observations, not theoretical constructs.

B) This is the correct answer. It accurately describes the difference between real-world and risk-neutral probabilities. Real-world probabilities incorporate uncertainties present in the financial market, while risk-neutral probabilities assume a risk-free environment without requiring additional returns.

C) This option incorrectly suggests that risk-neutral probabilities are more suitable for scenario analysis and stress testing, while real-world probabilities are appropriate for valuing financial instruments consistently with other market instruments. In reality, risk-neutral probabilities are used for consistent pricing of financial instruments, while real-world probabilities are favored for risk assessment and planning purposes.

D) This option incorrectly states that risk-neutral probabilities are lower than real-world probabilities due to the expected risk premium. In fact, risk-neutral probabilities are generally higher than real-world probabilities because they do not account for the risk premium demanded by investors in the actual market.

Q-18. Solution: C

A) This option describes judgmental approaches, not financial models. Judgmental approaches rely on expert judgment rather than theory-based principles.

B) This option describes empirical models, not financial models. Empirical models are data-driven and rely on historical data and statistical techniques.

C) This option correctly describes financial models. Financial models are theory-based and aim to describe the mechanics of the default process using fundamental economic principles.

D) This option also describes financial models, but it emphasizes the utilization of market data and sophistication for dynamic analysis, which is not the primary characteristic of financial models. Financial models are more concerned with fundamental economic principles and the mechanics of default.

Q-19. Solution: C

A) Incorrect. Internal credit rating systems used by banks focus on profitability ratios and balance sheet indicators rather than market information. They aim to assess creditworthiness based on factors like cash flow rather than relying solely on market data.

B) Incorrect. Rating agencies like Moody's and S&P aim for stability in their credit ratings and avoid frequent changes to prevent market disruptions. Frequent changes could lead to increased trading and transaction costs for bond traders.

C) Correct. Internal credit rating systems used by banks focus on profitability ratios and balance sheet indicators. It emphasizes the importance of cash flow in loan repayment, which aligns with the objectives of internal credit rating systems used by banks.

D) Incorrect. Moody's KMV and Kamakura are mentioned as companies providing estimates of a company's probability of default, but this does not suggest that they provide credit ratings for small and medium-sized companies without publicly traded debt.

Q-20. Solution: A

Option A accurately distinguishes between through-the-cycle (TTC) and point-in-time (PIT) credit rating systems. Through-the-cycle ratings are characterized by stability over an extended period, aiming to filter out short-term noise and fluctuations, while point-in-time ratings focus on short-term fluctuations and immediate credit risk, reflecting the borrower's current status. Options B, C, and D incorrectly describe the characteristics of TTC and PIT ratings, therefore are not the correct answer.

Q-21. Solution: A

A) The correct answer is A. The implementation phase involves mapping credit scores to risk rating classes based on empirical PD estimates and ensuring a well-diversified distribution of borrowers across rating categories. This step ensures that the derived ratings accurately represent the credit risk levels of borrowers.

B) This option is incorrect. While model validation does involve comparing the credit model's results to external assessments, such as ratings issued by credit rating agencies, this is not the primary focus of the validation process. The primary objective is to assess the expected predictive performance of the model in a context that resembles its actual use in practice.

C) This option is incorrect. Expert judgment is explicitly mentioned as a crucial aspect during the data collection and pre-processing phase.

D) This option is incorrect. All borrowers and loans should be re-rated at least annually, taking into consideration all new information about their status and progress. Continuous evaluation of borrowers is an essential aspect of the credit scoring and rating model development process.

Q-22. Solution: B

Statement I is false because the cumulative probability of default increases (i.e., even the highest rated companies will eventually fail over a long enough period). Statement II is true since the probability the firm survives over the next 6 years without default is: $(1-0.08)^6 = 60.6\%$.

Q-23. Solution: A

We first need to realize that the expected number of default in one year, which is $\lambda = 1/5 = 0.2$;
Using the Poisson distribution, we solve for the probability that x will be 1.

$$P(X = 1) = \frac{0.2^1 e^{-0.2}}{1!} = 16.37\%$$

Q-24. Solution: D**Q-25. Solution: A**

A is correct. The probability that the bond survives for 3 years and then defaults in Year 4 can be modeled as a Bernoulli trial given by the following equation, where FP stands for forward probability:

$$P(\text{default at end of Year 4}) = (1 - FP_1) \times (1 - FP_2) \times (1 - FP_3) \times FP_4 = 4.9\%$$

Q-26. Solution: C

$$PV(\text{risk-free debt}) - \text{put} = 80 \times e^{-0.04 \times 5} - 6.95 = 58.55$$

Q-27. Solution: A

When firms with subordinate debt are experiencing financial distress (low firm values), changes in the value of subordinate debt will react to changes in the model parameters in the same way as equity. Since equity is valued as a call option in the Merton model, a decline in interest rates will reduce the value of equity (and subordinate debt). When firms with subordinate debt are not experiencing financial distress (high firm values), changes in the value of subordinate debt will react to changes in the model parameters in the same way as senior debt. Since senior debt is valued as the difference in firm value less equity valued as a call option in the Merton model, a decline in interest rates will increase the value of senior debt and subordinate debt.

Q-28. Solution: D

A is wrong; the payoff of a bond holder is equivalent to a riskless bond minus a put on the value of a firm. B is wrong, the firm follows lognormal diffusion process, it doesn't allow for sudden change. C is wrong, because in this model default can only occur at the debt maturity.

Q-29. Solution: D

PD = $N(-d_2)$ where d_2 is the distance to default.

$$d_2 = \frac{\ln\left(\frac{20}{15}\right) + 12\% \times 1 - 0.5 \times 35\%^2 \times 1}{35\% \times \sqrt{1}} = 0.99$$

Such that $N(-d_2) = 16.11\%$

Please note we use the asset drift (12%) not the risk free rate, this is not option pricing!

Q-30. Solution: A

$$N[\ln(100/130) - (25\% - (30\%^2)/2) \times 5] / (30\% \times \text{sqrt}(5)) = 2.74\%$$

Q-31. Solution: A

The normal returns-based distance to default (DD) = $[\ln(400/300) + (14\% - 36\%^2/2) \times 1] / [36\% \times \text{sqrt}(1)] = 1.0080$

Q-32. Solution: B

Q-33. Solution: B

Just as an increase in the risk-free rate increases the value of a call option, an increase in the risk-free rate increases the equity value under Merton. However, the risk-free rate has no impact on the Merton PD; the actual return of 8% is used in that application.

Q-34. Solution: A

Distance to Default (DtD) approximates the number of standard deviations to reach the default threshold; thus, the higher the DTD, the least likely to default.

$$DtD = \frac{\ln V - \ln F + \left(\mu - \frac{\sigma^2}{2}\right)(t)}{\sigma\sqrt{t}}$$

DtD can be simplified by reducing the forward time periods to 1 ($t = 1$) and minimizing the drift factors $(\mu - \sigma^2/2)$ that tend to be small (assumed to equal 0) over one period to yield:

$$DtD = \frac{\ln V - \ln F}{\sigma}$$

Using this formula results in: Company P = $\ln(100/60)/0.10 = 5.11$

Company Q = $\ln(150/100)/0.07 = 5.79$

Company R = $\ln(250/160)/0.08 = 5.58$

Q is least likely to default; R is in the middle; P is most likely to default.

Q-35. Solution: B

Asset Value (V) Calculation:

The Merton model states:

$$E = V \times N(d_1) - D \times e^{-rT} \times N(d_2)$$

Plugging in the values and Solve for V:

$$50 = V \times 0.7 - 80 \times e^{-0.05 \times 2} \times 0.4$$
$$V \approx 112.8 \text{ million}$$

Asset Volatility (σ_V) Calculation:

Using the volatility relationship: $\sigma_E E = V \times N(d_1) \times \sigma_V$

Rearrange for σ_V :

$$\sigma_V = \frac{\sigma_E \times E}{V \times N(d_1)} \approx 25.3\%$$

Q-36. Solution: B

Moody's KMV model is a model for predicting private company defaults. It covers many geographic specific models, and each model reflects the unique lending, regulatory, and accounting practices of that region. Moody's KMV computes the normalized distance to default as:

$$DD = \frac{A - K}{\sigma_A}$$

where: "K" (floor) is defined as the value of all short term liabilities (one year and under) plus one half of the book value of all long term debt: 40 million + 0.5 × 20 million = 50 million. "A" is the value of assets: Market value of equity (1 million shares × 10/share = 10 million) plus the book value of all debt (60 million) = 70 million. Thus $A\sigma_A = 20\% \times 70 \text{ million} = 14 \text{ million}$.

$DD = (70 \text{ million} - 50 \text{ million}) / 14 \text{ million} = 1.429 \text{ standard deviations}$

Q-37. Solution: C

Default Point:

$$\left(500 + \frac{1}{2} \times 300\right) = 650$$

Distant to Default:

$$\frac{1200 - \left(500 + \frac{1}{2} \times 300\right)}{1200 \times 10\%} = 4.58$$

Q-38. Solution: D

While KMV computes the forward DD as a standard deviation, it does not assume normal returns/lognormal asset value. This is explicitly considered insufficient to model empirically heavy tails. Rather KMV first measures the distance-to-default as the number of standard deviations the asset value is away from default and then uses empirical data to determine the corresponding

default probability.

Q-39. Solution: A

T	Cumulative PD	Survival Probability	PD (t, t+1)	Conditional PD Given Survival Until Time t
1	$1 - e^{-0.15} = 0.1393$	$1 - 0.1393 = 0.8607$	0.1393	
2	$1 - e^{-0.15(2)} = 0.2592$	$1 - 0.2592 = 0.7408$	$0.2592 - 0.1393 = 0.1199$	$0.1199/0.8607 = 0.1393$

Q-40. Solution: C

Lambda = 6 defaults/12 month = 0.5 defaults per month.

$$P = 1 - \exp(-0.5) = 39.35\%$$

Q-41. Solution: D

The conditional one-year PD is equal to 8.6%, same as the unconditional one-year default probability.

Q-42. Solution: A

Using λ to represent the given hazard rate, we can calculate the cumulative probability of default in the first year using the formula $1 - \exp(-\lambda)$, which equals 0.09516.

Then, the cumulative probability that the firm defaults in the second year is equal to $1 - \exp(-2 \times \lambda)$ or 0.18127; Then, the probability of survival in the first year followed by a default in the second year is closest to: $0.18127 - 0.09516 = 0.08611$.

Q-43. Solution: B

Conditional default has a mean of $0.40 \times (-1) = -0.40$ and a volatility of $\sqrt{1 - 0.4^2} = 0.92$. The loss threshold is -2.33. Therefore the conditional default probability is:

$$\Phi\left(\frac{-2.33 - (-0.4)}{0.92}\right) = 1.8\%$$

Q-44. Solution: C

The asset return correlation of the two firms is then $\beta_1 \beta_2 = 0.5 \times 0.1 = 0.05$.

Q-45. Solution: A

Ratios	Ratios (A)	Model Coefficient (B)	A × B
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WC/TA	25.37%	1.2	0.304
RE/TA	44.78%	1.4	0.627
EBIT/TA	8.96%	3.3	0.296
EMV/TL	158.33%	0.6	0.950
S/TA	328.36%	0.999	3.280
Altman's Z-Score			5.46

Q-46. Solution: B

Q-47. Solution: B

Q-48. Solution: A

Selling a put option exposes you to zero counterparty credit risk as the premium is paid up front. The correct answer is therefore A. All the information necessary to price the option is provided but it is not necessary. The value of the put option is USD 24.90 (answer D) while the value of a call option with one year to expiration is 0.38.

Q-49. Solution: C

Q-50. Solution: D

The CD has the whole notional at risk. Otherwise, the next greatest exposure is for the forward currency contract and the interest rate swap. The short cap position has no exposure if the premium has been collected. Note that the question eliminates settlement risk for the forward contract.

Q-51. Solution: B

With an upward-sloping term structure, the fixed payer has greater credit exposure. He receives less initially, but receives more later. This back-loading of payments increases credit exposure. Conversely, if the forward curve flattens, the fixed payer (i.e., BNP Paribas) has less credit exposure. Credit Agricole must have greater credit exposure. Alternatively, if LIBOR drifts down, BNP will have to pay more, and its counterparty will have greater credit exposure.

Q-52. Solution: C

PFE of cross-currency swap, which combines the exposure of an interest rate swap and the FX forward. The dotted must be the average PFE (aka, AWCE) since it is a flat line.

Q-53. Solution: B

The risk of cross-currency swaps is driven by a large final payoff, and thus the profile increases monotonically until the maturity of the trade. The FX risk of the notional exchange dominates the small contribution due to interest rate exposure.

Q-54. Solution: A

A is correct. Given the risk-free rate of 3.0%, we can estimate the forward price (at maturity, in nine months) of the contract as:

$$\text{Forward price} = \text{spot} \times e^{r \times t} = 92 \times e^{0.03 \times 0.75} = \text{EUR } 94.093 \text{ million}$$

Today, after 6 months (3 months to maturity)

$$\text{Forward contract price estimate} = 94.093 \times e^{-0.03 \times 0.25} = \text{EUR } 93.39 \text{ million}$$

Note that, Forward Contract Value = Credit Risk Exposure;

Therefore, given that the current (with 3 months remaining to maturity) underlying asset price of EUR 94 million, the long forward contract's value is given by:

$$\text{Current value} = 94 - 93.39 = \text{EUR } 0.61 \text{ million}$$

which represents exposure, and is the value to the long (TMI) because the contract is a claim on the asset, which is currently worth EUR 94.0 million, and an obligation to pay EUR 94.093 million for it in 3 months. Because the contract value of EUR 0.610 million is positive, the long counterparty (TMI) bears the credit risk exposure.

$$\text{Positive exposure} = \text{Max}(\text{value}, 0)$$

$$\text{Negative exposure} = \text{Min}(\text{value}, 0)$$

For forwards, while there is no current exposure (because payment is only made at expiration, there is always positive potential exposure as long as market price > contract price, and negative potential exposure if market price < contract price. At origination (time 0), there is neither current nor potential exposure (since market price = contract price).

B is incorrect (see explanation above).

C and D are incorrect. They compute the contract price incorrectly by discounting the forward value over 6 months and not 3 months.

Q-55. Solution: B

Understanding which trades have the greatest impact on a counterparty's credit value adjustment requires use of the marginal CVA. Incremental CVA, by contrast, is useful for pricing a new trade with respect to an existing one.

Q-56. Solution: D

From the perspective of the financial institution:

$-EPE \times \text{counterparty credit spread} - ENE \times \text{institution credit spread}$

$= -5\% \times 300 - (-3\%) \times 200 = -9 \text{ bps}$

This is what the financial institution may charge the counterparty for overall counterparty risk.

Q-57. Solution: A

Q-58. Solution: C

The BSM call option price $= 100 \times 0.64 - 100 \times e^{-3\%} \times 0.4 = \25.182

which does not include counterparty risk incurred by the long option position (the short has no counterparty risk). The CVA-adjusted value $= \$25.182 - \$23.00 \times 5\% \times 75\% = \24.32

Q-59. Solution: C

We refer to wrong-way risk as the adverse (negative) correlation between the exposure to the counterparty and its credit quality. Alternatively, it can be stated as the positive correlation between exposure and credit spread.

Q-60. Solution: C

Q-61. Solution: B

The BCVA formula differs from the CVA formula in that BCVA incorporates expected negative exposure (ENE), and the probability of the counterparty's survival must be included in the BCVA formula.

Q-62. Solution: C

The lower credit qualities and increased credit spreads should result in higher DVA and CVA for both ADB and HIP. Therefore, only C is correct and A, B and D are all incorrect.

Q-63. Solution: A

A is correct. To derive the credit valuation adjustment (CVA), we use the standard formula:

$$CVA = - \sum_{i=0}^n LGD_i \times EPE_i \times PD_i \times DF_i$$

Where (at any time f),

The discount factor (DF_i) is determined from the risk-free rate of 2%; and

$$\text{Hazard rate} = \frac{\text{Spread}}{1-\text{RR}} = 12\% \text{ (True for years 2 and 3)}$$

The probability of default PD(t), marginal probability, is derived from the relationship between cumulative probability of default PD(c,t) and the constant hazard rate (X):

$$\text{PD}(c, t) = 1 - e^{-\lambda t}$$

$$\text{PD}(c, 1) = 1 - e^{-0.12 \times 1} = 11.31\%, \text{ and so } \text{PD}_1 = 11.31\%$$

$$\text{PD}(c, 2) = 1 - e^{-0.12 \times 2} = 21.34\%, \text{ and so } \text{PD}_2 = 21.34\% - 11.31\% = 10.03\%$$

$$\text{PD}(c, 3) = 1 - e^{-0.12 \times 3} = 30.23\%, \text{ and so } \text{PD}_3 = 30.23\% - 21.34\% = 8.89\%$$

	Year 0	Year 1	Year 2	Year 3
Hazard Rate (λ)		12%	12%	12%
Probability of default (cumulative) (PD(c, t))		11.31%	21.34%	30.23%
Probability of default (marginal) (PD(t))		11.31%	10.03%	8.89%
Discount factor (DF)		0.9804	0.9612	0.9423
Recovery rate		0.85	0.75	0.65
Expected positive exposure (EPE) (AUD million)		15	15	15
Collateral (C) (AUD million)		13	13	13
EPE' (netted) (AUD million)		2	2	2
LGD $\times$ EPE' $\times$ PD(t) $\times$ DF		0.0333	0.0482	0.0586
CVA (AUD million)	-0.1401			

B is incorrect. -AUD 0.172 million is the result obtained when the hazard rate is used instead of the PD. C is incorrect. -AUD 0.442 million is the result obtained when the recovery rate and not the LGD is used. D is incorrect. -AUD 1.051 million is the result obtained when collateral is not considered.

Q-64. Solution: C

$$\text{Stress Loss} = \text{Stressed EL} - \text{EL}$$

$$= \text{Stressed PD} \times \text{average EPE} \times \text{LGD} - \text{PD} \times \text{average EPE} \times \text{LGD}$$

$$= (1.60\% - 1.00\%) \times 300 \times 65\% = 1.17 \text{ million}$$

Q-65. Solution: B

B is correct. Let p denote Bank Gamma as the party (as the party making the calculation), and c denote Bank Phi as counterparty. Also, note that the BCVA = CVA + DVA. So,

$$\text{CVA} = -\text{LGD}_c \times \text{EPE}_p \times \text{PD}_c \times (1 - \text{PD}_p) = -(0.08 \times 60,000,000 \times 0.018 \times 0.975) = \text{CNY } -84,240$$

$$\text{DVA} = -\text{LGD}_p \times \text{ENE}_c \times \text{PD}_p \times (1 - \text{PD}_c) = -(0.18 \times -45,000,000 \times 0.025 \times 0.982) = \text{CNY } 198,855$$

$$\text{Therefore, BCVA} = \text{CVA} + \text{DVA} = \text{CNY } -84,240 + \text{CNY } 198,855 = \text{CNY } 114,615$$

A is incorrect. CNY 84,240 is the CVA of Bank Gamma and the negative sign is ignored.

C is incorrect. CNY 198,855 is the DVA of Bank Gamma.

D is incorrect. CNY 201,960 is the result obtained when the expected positive exposures in the CVA and the DVA formulas are switched and ignored the sign of BCVA.

Q-66. Solution: B

1. CVA Calculation:

$$CVA = \text{Spread}_{CP} \times EPE$$

During stress: Spread_{CP} increases (due to wider credit spreads). EPE increases (given in the scenario). Both factors drive CVA to increase.

2. DVA Calculation:

$$DVA = \text{Spread}_{FI} \times NEE$$

During stress: Spread_{FI} increases (due to wider FI credit spreads). ENE remains unchanged (given in the scenario). The two factors increases DVA.

Q-67. Solution: C

Q-68. Solution: D

The hedge fund has wrong-way risk. As interest rates rise, both Bank HJK's and Bank PQR's equity value would decline since the performances of the two banks are highly correlated. Therefore, the value of the long put option on PQR would increase, resulting in a higher exposure to bank HJK for the hedge fund. This is a wrong-way risk since the hedge fund's exposure to HJK would be increasing as the credit quality of HJK is declining.

The manufacturer also has wrong-way risk. Since the credit spread of Bank HJK is increasing and credit spreads of different banks in the same market tend to be positively correlated, the credit spread of Bank PQR should also increase. Therefore, the value of the manufacturer's long CDS position on Bank PQR is increasing at the same time the credit quality of Bank HJK is decreasing; thus, that is wrong-way risk.

Q-69. Solution: D

D is correct. "Buying a put option on a stock (or stock index) where the underlying in question has fortunes that are highly correlated to those of the counterparty is an obvious case of wrong-way risk". Thus, choices A and C are ruled out.

Also, "an out-of-the-money put option will have more wrong-way risk than an in-the-money one." That implies option D is the only correct one.

Q-70. Solution: C

CCPs are at risk of wrong-way risk (WWR) because they disassociate credit quality and exposure. In regard to (A), (B) and (D), each is FALSE.

Q-71. Solution: D

D is correct. The CCP's net exposure is AUD 28 million under central clearing. We express the positions – through the process of novation and netting – and present the results and the explanations below (note that a negative net exposure means a zero exposure):

Firm 1's exposure to CCP = $90 + 12 = \text{AUD } 102 \text{ million}$; CCP's exposure to Firm 1 = $60 + 70 = \text{AUD } 130 \text{ million}$.

Hence, Firm 1's exposure to CCP = $\text{AUD } 0 \text{ million}$ ($= 102\text{m} - 130\text{m} = -28\text{m} = 0$).

Firm 3's exposure to CCP = $\text{AUD } 70 \text{ million}$; CCP's exposure to Firm 3 = $12 + 57 = \text{AUD } 69 \text{ million}$.

Hence, Firm 3's exposure to CCP = $\text{AUD } 1 \text{ million}$ ($= 70\text{m} - 69\text{m}$).

Firm 2's exposure to CCP = $60 + 57 = \text{AUD } 117 \text{ million}$; CCP's exposure to Firm 2 = $\text{AUD } 90 \text{ million}$.

Hence, CCP's exposure to Firm 2 = $\text{AUD } 27 \text{ million}$ ($= 117\text{m} - 90\text{m}$).

Q-72. Solution: B

Netting means that the payments between the two counterparties are netted out, so that only a net payment has to be made. With netting, the investment firm is not required to make every payout, hence the loss will be reduced to: $\text{USD } 32 \text{ million} + \text{USD } 12 \text{ million} - \text{USD } 16 \text{ million} - \text{USD } 8 \text{ million} = \text{USD } 20 \text{ million}$. Without netting, the loss is the outstanding long position: $\text{USD } 32 \text{ million} + \text{USD } 12 \text{ million} = \text{USD } 44 \text{ million}$.

Q-73. Solution: A

The properly netted amounts are: For A: $B = \$0$, $C = \$5$, $D = \$0$, for a sum of $\$5$. For B: $A = \$0$, $C = \$0$, $D = \$8$, for a sum of $\$8$. For C: $A = \$0$, $B = \$5$, $D = \$1$, for a sum of $\$6$. For D: $A = \$9$, $B = \$0$, $C = \$0$, for a sum of $\$9$.

Q-74. Solution: D

The greatest netting benefit among the scenarios presented occurs when the two trades have a strong negative correlation. In this case, a large portion of the negative exposures will offset positive exposures.

Q-75. Solution: C

Define X and Y as the absolute values of the positive and negative positions. The net value is $X - Y = 20 \text{ million}$. The absolute gross value is $X + Y = 80$. Solving, we get $X = 50 \text{ million}$. This is the positive part of the positions, or exposure.

Q-76. Solution: D

“a” is a walk-away clause.

“b” is a close-out netting.

“c” is Termination.

Q-77. Solution: D

Increasing collateral would effectively reduce current credit exposure depending on the contract parameters, mainly minimum transfer amount and threshold.

Q-78. Solution: A

A is correct. Threshold is the amount of uncollateralized exposure. A lower threshold value means a larger portion of exposure is protected by collateral.

Similarly, C is incorrect because a lower independent amount is equivalent to a higher threshold. That corresponds to a smaller portion of exposure being protected.

B is incorrect because the margin period of risk is the effective time assumed between a collateral call and receiving the appropriate collateral. It has nothing to do with the amount of collateral.

D is incorrect. Collateral has little effect at both the beginning and the end of the exposure profile when the exposure is relatively small.

Q-79. Solution: B

The overall annual volatility of the position (exposure + collateral) is $(0.06^2 + 0.08^2) = 0.10$ or 10%. Therefore, actual calculations of the changes in value of the position for a specified time horizon and confidence level are not necessary, since 10% is greater than that of the exposure (6%) and the collateral (8%). The correct choice is B – the collateral mitigates the exposure today while increasing the volatility of the position in the future.

Q-80. Solution: A

Counterparty exposure, in theory, can be almost completely neutralized as long as a sufficient amount of high quality collateral, such as cash or short-term investment grade government bonds, is held against it. If the counterparty were to default, the holder of an open derivative contract with exposure to that counterparty would be allowed to receive the collateral. Cross-product netting would only reduce the exposure to one of the counterparties, and purchasing credit derivatives would replace the counterparty risk from the individual counterparties with counterparty risk from the institution who wrote the CDS.

Q-81. Solution: C

Both can incur counterparty risk. In regard to (A), (B) and (D), each is TRUE.

Q-82. Solution: D

If it uses a clearinghouse and the clearinghouse makes a mistake (operational risk) like that made by ABC, ABC will have recourse to the clearinghouse and it would have, therefore, reduced its operational risk exposure.

A. Incorrect. ABC is not funding from Repo Co.

B. Incorrect. The use of a clearinghouse does not change Repo Co.'s default risk – just ABC's exposure to Repo Co. defaults.

C. Incorrect. The use of a clearinghouse in this situation does not reduce XYZ's lending risk.

Q-83. Solution: C

Payment netting is the simple netting of cash flows due on the same day.

Closeout netting occurs if there is an event of default, which would include an incidence of fraud.

One of the shortcomings of clearinghouses, and closeout netting as well, is that the other party, in this case ABC, jumps to the head of the queue with its claim on Repo Co. to the possible detriment of others, particularly those outside the clearinghouse in general.

Thus, only C is correct.

Q-84. Solution: A

A is correct.

Additional collateral (C) required for posting can be explained from the mark-to-market value of collateral posted (X), mark-to-market value of net exposure (E), the threshold (K), and the minimum transfer amount (MTA) as follows:

(i) Collateral call (C) can be made if: $E > (K + MTA + X)$

(ii) The collateral amount required: $C = E - K - X$, and the amount is positive if $(E - K - X) > MTA$, otherwise it is zero.

In this example:

$(K + MTA + X) = 14,000,000 + 2,500,000 + 10,800,000 = 27,300,000 > E = 27,000,000$ which corresponds to no collateral call. Thus, A is correct.

B is incorrect. $CNY\ 1,990,000 = \text{new exposure} - \text{original exposure} - \text{rounding amount} = 27,000,000 - 25,000,000 - 10,000$, which is incorrect.

C is incorrect. $CNY\ 2,000,000$ is the difference between the new net exposure and the original net exposure ($= 27,000,000 - 25,000,000 = CNY\ 2,000,000$).

D is incorrect. CNY 2,500,000 is the minimum transfer amount.

Q-85. Solution: C

The first layer in the loss waterfall is for the defaulting member's collateral and default fund contributions to be exhausted. The next layer is for the CCP to tap into its own equity to the point where it could still function normally. Non-defaulting members will then have their default funds exhausted before moving to the rights of assessment.

Q-86. Solution: C

CCPs reduce counterparty risk, but they do not eliminate this risk. Improved legal and operational efficiencies, increased transparency, and loss mutualization are major advantages of the CCP central clearing process.

Q-87. Solution: A

Option A accurately summarizes the key difference between initial margin and default fund requirements. Initial margin aims to provide coverage with a high level of confidence against losses, while default funds are utilized to absorb extreme losses beyond initial margin through mutualization, which ultimately makes clearing cost-effective but may potentially increase moral hazard. Option B incorrectly suggests that initial margin requirements are significantly larger than default funds, which is not necessarily true. Option C provides accurate information about the role of initial margin but does not emphasize the difference between them. Option D is inaccurate in stating that default funds are updated more frequently than initial margins; in fact, default fund contributions are not updated as frequently as initial margins, typically on a monthly basis.

Q-88. Solution: B

Loss Absorption Steps:

Step 1: Initial Margin from the defaulting member = 10.

Step 2: Default Fund from the defaulting member = 20.

Step 3: CCP's Skin-in-the-Game Equity = 25.

✓ Total absorbed so far = $10 + 20 + 25 = 55$.

✓ Remaining loss = $75 - 55 = 20$.

Step 4: Remaining loss is absorbed by non-defaulting members' Default Fund.

✓ Number of non-defaulting members = $10 - 1 = 9$.

✓ Total Default Fund from non-defaulting members = $9 \times 20 = 180$.

✓ Loss allocated to non-defaulting DF = 20. Remaining Default Fund Reserve = $180 - 20 = 160$.

Q-89. Solution: D

The protection buyer is exposed to the joint risk of default by the counterparty and underlying credit. If only one defaults, there is no credit risk.

Q-90. Solution: A

Because LIBOR is flat, the fixed-coupon yield is also 4.6%, creating a spread of $800 - 460 = 340\text{bp}$ on the bond. Going long the bond and short credit via buying the CDS yields an annual profit of $340 - 150 = 190\text{bp}$.

Q-91. Solution: A

A break clause – a.k.a., mutual put – is an agreement to terminate (break) a transaction at pre-specified dates in the future.

Q-92. Solution: B

B is correct. In the trade compression, only contracts on the same reference credit can be compressed, and so the bank's long position with Digital Corporation on WKL Company as reference credit is not considered.

Therefore,

Net contract notional value = Long 9,000,000 + Short 7,000,000 = Long 2,000,000

A is incorrect. Long EUR 1 million is the result of netting the trades with the same coupon and incorrectly stating it as a net long instead of a net short position.

C is incorrect. Long EUR 8 million is the result of netting all the trades.

D is incorrect. Long EUR 15 million is the result of summing the positions with the same counterparty.

Q-93. Solution: A

Option A correctly defines the CDS-bond basis as the difference between the CDS spread and the bond yield spread, highlighting factors that affect its deviation from zero, including market pricing discrepancies, counterparty risk, and accrued interest exclusion. Option B incorrectly suggests a correlation between the spreads without addressing the basis itself or its influencing factors. Option C discusses historical trends but doesn't accurately define the basis or its influencing factors. Option D confuses the CDS-bond basis with the estimation of default probabilities using credit spreads and recovery rates, which is a related but distinct concept. Therefore, option A is the correct answer.

Q-94. Solution: B

Option B accurately reflects the role of credit indices and fixed coupons in pricing CDS transactions. Credit indices streamline trading by providing an average CDS spread for a group of companies within a portfolio, promoting efficiency and simplicity. Fixed coupons ensure consistent payments over the contract period, contributing to predictability and stability in cash flows for both parties involved.

Q-95. Solution: A

Q-96. Solution: B

The NASDAQ will increase 25%, or $(3625/2900) - 1$, over the next year, so the index payer will pay USD 5 million (0.25×20 million) to ABC Corp. Since ABC Corp's payments depend on today's LIBOR, it will pay 3.75% ($1.25\% + 2.5\%$) or USD 0.75 (0.0375×20 million). So the firm's net cash flow would be 0.75 million – 5 million = -USD 4.25 million.

Q-97. Solution: D

The TRS provides a hedge against credit and market risk. In the case of LIBOR, a higher interest rate implies a loss on the reference, however this is hedged by the protection sellers higher payments.

Q-98. Solution: A

Option A: This option correctly explains that the Gaussian copula model is first used to calculate the joint default probabilities, which are later utilized in the valuation process through the spread payments approach to discount future expected cash flows to their present values.

Option B: The description of this option is incorrect. The Gaussian copula model is used to calculate joint default probabilities, not individual default intensities, and the spread payments approach is used to discount future cash flows, not to calculate them.

Option C: The description of this option is incorrect. The Gaussian copula model does not predict time to default; rather, it models the dependence structure between defaults. Also, the spread payments approach does not estimate fair spreads based on historical data but discounts future expected cash flows.

Option D: This choice confuses the two methods. The spread payments approach is about discounting cash flows given a certain risk profile, not establishing an overall risk premium. While the Gaussian copula model can indeed simulate correlated defaults, it doesn't directly reallocate premiums among tranches.

Q-99. Solution: B

B is correct. the weighted average cost of the securitization may be lower than the cost of the bank's current debt. This is often the case if the credit quality of the securitized assets is higher than the credit quality of the bank's balance sheet as a whole.

A is incorrect. The statement incorrectly describes overcollateralization, which is an enhancement made to the assets that raises their credit quality. Specifically, the credit quality of the securitized car loan assets would be raised during the credit rating process if the principal value of notes issued is lower (not higher) than the principal value of the assets.

C is incorrect. The statement describes an amortizing (pass-through) SPV structure and not a revolving SPV structure, which can provide principal payments in equal installments or as lump sum payments at maturity. However, it incorrectly states that pre-payment assumptions are not incorporated.

D is incorrect. The first part of the statement describes the master trust SPV structure, not the revolving structure, and the reference to weighted-average life (WAL) describes the amortization structure.

Q-100. Solution: B

The senior tranche will gain value if the default correlation decreases. High correlation implies that if one name defaults, a large number of other names in the CDO will also default. Low correlation implies that if one name defaults, there would be little impact on the default probability of the other names. Therefore, as the correlation decreases, the cumulative probability of enough defaults occurring to exceed the credit enhancement on the senior tranche will also decrease. Hence the investor who has sold protection on the senior tranche will see a gain.

Q-101. Solution: B

USD 50 million is calculated by $\text{USD}300 - \text{USD}250 = \text{USD}50$, since prepayments are first distributed to the senior tranches.

Q-102. Solution: D

An investor's claim when purchasing an ABS is solely with the ABS and no longer with the originator. The fact that KMAC has \$10 million set aside means nothing for the ABS issue if it remains on KMAC's balance sheet and is not part of the ABS issue. The other answer choices all describe forms of credit support that will support at least Tranches X and A, if not all 3 tranches. By having Tranche Z be subordinate to Tranche X, Tranche X has additional support. Also, loans of \$125 million are used to back asset-backed securities worth $(\$50 + \$35 + \$35) = \120 million, which means the issue, is over-collateralized. The weighted average interest rate paid on the securities is approximately

6.2%. If the weighted average interest rate on the loans that make up the pool is 6.4% that means there is an excess spread between the loans and securities that also provides support for the entire issue.

Q-103. Solution: D

Excess spread = 100 million $\times$ (LIBOR + 200 bps – 20bps) – 90 million $\times$ (LIBOR + 100bps)
= 10 million $\times$ (LIBOR + 9%)

Q-104. Solution: D

Q-105. Solution: D

Covered bond and securitization are similar in the sense that both methods do issue securities, which are backed by assets, as a means to raise funds (borrow) in capital markets.

Q-106. Solution: A

Q-107. Solution: C

The total value of the tranches is: \$400 + \$120 + \$50 = \$570 million. The value of the collateral is \$600 million, so the CMO is over collateralized by \$30 million.

The net excess spread is 8.75% - 7.50% - 0.60% = 0.65%, so there is positive excess spread. This provides credit enhancement for the CMO investors.

Q-108. Solution: A

Internal enhancements include: overcollateralization, direct equity issue, holdback, cash collateral account (CCA), excess spread.

External credit enhancement include, insurance, warps, and guaranties, letters of credit, CDS, put options on assets.

Q-109. Solution: B

Q-110. Solution: C

The default risk in a CDO is structured through various tranches in such a way that a pool of assets that were once lower rated could be AAA rated after the securitization process. The equity tranche is the most junior tranche. Therefore, it offers the highest return potential but with no certain return. The equity tranche also bears the highest level of default risk.

Q-111. Solution: B

Revolving structures are used with products that are paid back on a revolving basis, such as credit card debt or auto loans. Credit card debt does not have a pre-specified amortization schedule; therefore the principal paid back to investors is in large lump sums rather than amortizing schedules.

Q-112. Solution: B

$CPR = 0.4\%$, therefore, $SMM = 1 - (1 - CPR)^{(1/12)} = 0.03\%$

$I/Y = 5.5/12$, $N = 60$, $PV = 20,000,000$, $FV = 0 \rightarrow PMT = 382,023.24$

Interest payment = $20,000,000 \times 5.5\%/12 = 91,666.67$

Principal payment = $382,023.24 - 91,666.67 = 290,356.58$

$SMM = \text{prepayment}/(\text{beginning balance} - \text{scheduled principal payment}) = 0.03\% = \text{prepayment}/(20,000,000 - 290,356.58)$

Therefore, prepayment = 6,581.96

Q-113. Solution: D

The WAM is calculated as follows: $WAC = [90(2 \text{ million}) + 180(3 \text{ million}) + 270(5 \text{ million}) + 360(10 \text{ million})]/(2 \text{ million} + 3 \text{ million} + 5 \text{ million} + 10 \text{ million}) = 284 \text{ days}$

Q-114. Solution: C

C is correct: The value of the overcollateralization account (OC) at end of year 2 (OC2) includes the compounded year-1 value of the OC (OC1), the recovered principal amount at the end of year 2 (R2) and is computed as: $OC2 = (1 + \text{swap rate}) * OC1 + R2$.

Since there is no default in year 1, consider: $\text{Min}[(0.035 + 0.022) * 100 * 1,000,000 - 6,300,000; 1,500,000] = -600,000$

As that is negative, the overcollateralization account is entirely depleted at the end of year 1. For year 2,

First: Excess spread at the end of year 2 = $L - B$, where,

$L = \text{interest from surviving loans} = (0.035 + 0.022) * (100 - 8) * 1,000,000 = \text{EUR } 5,244,000$

$B = \text{bond coupon interest due to both junior and senior bonds} = \text{USD } 6,300,000$ (given)

Therefore, excess spread = $5,244,000 - 6,300,000 = \text{EUR } -1,056,000$ (negative).

Second: Recovery principal amount in year 2 = $R2 = 0.45 * 8 * 1,000,000 = \text{EUR } 3,600,000$, which would flow into the overcollateralization account (OC). Therefore, since the OC had a net EUR 0 at the beginning of year 2, then the interest shortfall (calculated in the first step above, as negative) should be paid from the OC account, leaving the OC with a net amount = $3,600,000 - 1,056,000 =$

EUR 2,544,000.

Q-115. Solution: C

2023 OC Account Calculation:

- Interest Income: $50 \text{ million} \times 8.1\% = 4.05 \text{ million}$.
- Senior/Mezzanine Interest Paid:
 - ✓ Senior: $32 \text{ million} \times 6.1\% = 1.952 \text{ million}$.
 - ✓ Mezzanine: $11 \text{ million} \times 9\% = 0.99 \text{ million}$.
 - ✓ Total = $1.952 \text{ M} + 0.99\text{M} = \2.942 million .
- Excess Interest: $4.05 \text{ M} - 2.942\text{M} = \1.108 million .
- Capped Inflow to OC: \$0.5 million (due to 2023 maximum inflow rule).
- 2023 OC Balance: $0 + 0.5\text{M} = \$0.5 \text{ million}$.

2024 OC Account Calculation:

- Defaults: 6 loans $\rightarrow$ \$3.75 million principal defaulted.
- Recoveries: $3.75 \text{ M} \times 46\% = 1.725 \text{ million}$ (added to OC without cap).
- Remaining Loans: 74 loans $\rightarrow$ \$46.25 million.
- Interest Income: $46.25 \text{ M} \times 8.1\% = 3.746 \text{ million}$.
- Senior/Mezzanine Interest Paid: \$2.942 million (same as 2023).
- Excess Interest: $3.746 \text{ M} - 2.942\text{M} = \0.804 million .
- Capped Inflow to OC: \$0.75 million (due to 2024 maximum inflow rule).
- Total OC Balance (2024):
 - ✓ Previous balance: $\$0.5 \text{ M} \times (1+5.25\%) = 0.52625 \text{ million}$.
 - ✓ Add 2024 inflows: $+ 0.75\text{M}$ (capped excess interest) $+ 1.725\text{M}$ (recoveries).
 - ✓ Total = $0.52625\text{M} + 0.75\text{M} + 1.725 \text{ M} = 3.00125 \text{ million}$.

Q-116. Solution: B

The loan-to-value ratio represents the amount of the mortgage versus the appraised value of the property. The higher this ratio is for a property and an associated borrower, the more risk there is to the lender. In order to protect their position, a lender will charge a higher interest rate. Each of the other scenarios will result in a lower interest rate.

Q-117. Solution: A

LTV = 80%

$\text{Loan1} = 80\% \times 500,000 = 400,000$

$\text{Monthly payment1} = 400,000 \times 2.75\%/12 = 917$

$\text{Loan2} = 80\% \times 400,000 = 320,000$

$I/Y = 4.5/12$, $N = 15 \times 12$, $PV = 320,000$, $FV = 0 \rightarrow PMT = 2,448$

Change in monthly payment = $2,448 - 917 = 1,531$

Q-118. Solution: B

Q-119. Solution: C

C is correct. The dark side of retail credit portfolios are given by a sudden rise of credit losses due to unexpected but systematic factors that influence the creditworthiness and the behavior of many credit customers in a bank's retail portfolio. An example of a systematic risk factor is the weak (or undetected) performance or the bias of credit risk assessment tools for retail customers.

A is incorrect. Focusing on big corporations does not address the dark side risk affecting the retail credit.

B is incorrect. The systematic risk factors mentioned are non-diversifiable risks.

D is incorrect. Stress tests only help to evaluate/determine risk but do not mitigate risks.

Q-120. Solution: C

A) This statement is incorrect. Foreign currency defaults occur due to shortages of foreign currency, not constraints on printing currency. Local currency defaults can happen when countries lack the ability to print more currency, but it's not the primary reason.

B) This statement is incorrect. Foreign currency defaults involve debt denominated in a foreign currency, not the country's own currency. Local currency defaults occur due to shortages of the country's own currency, not the foreign currency.

C) This statement is correct. Local currency defaults can indeed happen even when countries have the ability to print more currency, while foreign currency defaults are often driven by deliberate trade-offs between default and currency debasement.

D) This statement is incorrect. Local currency defaults occur due to shortages of the country's own currency, not foreign currency shortages. Foreign currency defaults occur precisely because countries lack the ability to print more foreign currency.

Q-121. Solution: B

A) This statement is incorrect because although credit ratings are widely recognized, they are slow to update and lack granularity compared to market-based measures. Additionally, they are subject to potential bias.

C) This statement is incorrect because while CDS spreads do offer real-time updates, they are susceptible to counterparty and liquidity risks, which may affect their accuracy in predicting default.

D) This statement is incorrect because market-based credit default spreads are indeed influenced by factors unrelated to default risk, such as liquidity and investor demand, making them less reliable predictors of default compared to credit ratings and CDS spreads.

Q-122. Solution: C

- A. This statement is incorrect. Ratings are often upward biased rather than overly pessimistic.
- B. This statement is incorrect. There is herd behavior among ratings agencies, suggesting that their assessments are not always independent.
- C. This statement correctly identifies a shortcoming mentioned. Ratings agencies often take too long to update sovereign ratings, leaving investors unprotected during crises.
- D. This statement is partially correct but does not accurately capture the essence of the issue. While biased assessments are mentioned, the bias is not solely attributed to government influence, but also to factors such as limited resources and revenue bias.

Q-123. Solution: C

- A) This option is incorrect. Guidelines are meant to facilitate compliance with credit risk management standards, not serve as punitive measures for breaches.
- B) This option is incorrect. Guidelines are intended to provide clear rules for approving transactions involving credit risk, ensuring thorough examination rather than swift approval.
- C) This option is correct. Guidelines play a crucial role in establishing clear rules for approving transactions involving credit risk, ensuring compliance across the organization.
- D) This option is incorrect. Guidelines are designed to be concise, precise, and accessible, not complex documents which may confuse individuals.

Q-124. Solution: C

- A) Incorrect. The credit committee considers various disciplines, not just financial aspects.
- B) Incorrect. The credit committee comprises individuals from various disciplines to ensure diverse perspectives are considered.
- C) Correct. This aligns with the role described: ensuring thorough evaluation before approval.
- D) Incorrect. The credit committee is chaired by a respected individual who maintains a neutral and objective environment during deliberations, not necessarily a member of the business unit.

Q-125. Solution: C

In this scenario, all the approval documents were appropriately distributed to the committee before the meeting. Sarah did spend time explaining the investment to the committee, but she also helped facilitate discussion among the committee members and listened to everyone's opinion. Votes do not have to be unanimous, but they should be held unless the decision is unanimous. However, meetings should be accurately documented in case an investment goes awry. In the case of a bad transaction, auditors would want to know what was discussed in the meeting and if the reasoning is behind the investment decision.

Q-126. Solution: D

Approvals should be additive meaning each authority level along the way must approve the transaction. Therefore, before the investment could make it to the credit committee, both the head of trading and the head of risk would have had to sign off on the investment.

Q-127. Solution: C

ABC decided to retain the loss assets on the balance sheet, which inflated its assets and loan loss reserves compared to XYZ. If James believes the loan will not be recovered, the best way to get the firms on a comparable basis is to take the loan off ABC's balance sheet by reducing its assets and loan loss reserves. Conversely, if James believed that the loan could one day be recovered, he would want to increase XYZ's assets on its balance sheet and loan loss reserves.

Q-128. Solution: B

- A) This statement is incorrect because defining exposure quantification is a challenge, and contingent liabilities, including guarantees and letters of credit, should be included within the exposure limit.
- B) This statement is correct because establishing limits for related parties is typically stipulated by regulators, but if not, banks should maintain such limits as a matter of board policy.
- C) This statement is incorrect because modern prudential regulations usually stipulate that a bank refrain from investing in or extending credit to any individual entity or related group of entities in excess of a prescribed percentage of the bank's capital and reserves.
- D) This statement is incorrect because the treatment of collateral is a contentious issue due to the subjective nature of collateral valuation, and it should not be considered when determining the size of exposure.

Q-129. Solution: D

- A) This option is contrary to the description in the core reading. According to the core reading, larger banks tend to decentralize based on geographical area, lending products, and customer types, while in smaller banks, it is typically centralized.
- B) This option contradicts the advice given in the core reading. The core reading states that a lending policy should establish limits for all lending officers to ensure proper control over loan approval.
- C) This option contradicts the advice given in the core reading. The core reading indicates that prudent banking practice requires board approval for all facilities extended to related parties.
- D) This option aligns with the description in the core reading. According to the core reading, a lending policy should provide an outline of the scope and allocation of credit facilities and the management of the credit portfolio, including the origination, appraisal, approval, disbursement, monitoring, collection, and handling procedures for various credit functions.